

Collaborative Ocular Melanoma Study (COMS) Randomized Trial of I-125 Brachytherapy for Medium Choroidal Melanoma

I. Visual Acuity after 3 Years COMS Report No. 16

Collaborative Ocular Melanoma Study Group

Objective: To report visual acuity during the first three years after iodine 125 (I^{125}) brachytherapy for medium-sized choroidal melanoma and to identify important baseline and treatment factors associated with posttreatment visual acuity in a group of patients who were treated and observed prospectively as part of a large, randomized clinical trial.

Design: Observational case series within a randomized, multicenter study.

Participants: Patients enrolled in the Collaborative Ocular Melanoma Study randomized trial of I^{125} brachytherapy versus enucleation had choroidal melanoma of at least 2.5 mm but no more than 10.0 mm in apical height, and no more than 16.0 mm in largest basal dimension. One thousand three hundred seventeen patients enrolled from February 1987 through July 1998; 657 patients were assigned to I^{125} brachytherapy. Visual acuity data for 623 patients who received I^{125} brachytherapy as randomly assigned and who have been observed for at least 1 year were analyzed for this report.

Methods: Under study protocol, an ophthalmic evaluation, including best-corrected visual acuity measurement of each eye, was performed at baseline, every 6 months thereafter for 5 years, and once yearly thereafter. Two poor vision outcomes, visual acuity of 20/200 or worse that was confirmed at the next follow-up examination and loss of six lines or more of visual acuity from baseline that was confirmed at the next follow-up examination, were analyzed to identify baseline and treatment characteristics that were associated with posttreatment visual acuity.

Results: At baseline, median visual acuity in the eye with choroidal melanoma was 20/32, with 70% of eyes having 20/40 or better and 10% of eyes having 20/200 or worse visual acuity. Three years after I^{125} brachytherapy, median visual acuity was 20/125, with 34% having 20/40 or better and 45% having 20/200 or worse visual acuity, including eyes that were enucleated within 3 years of treatment. Life-table estimates of percentages of patients who lost six or more lines of visual acuity from baseline, a quadrupling of the minimum angle of resolution, with this finding confirmed at the next 6-month follow-up examination, were 18% by 1 year, 34% by 2 years, and 49% by 3 years after treatment. Life-table estimates of percentages of patients with baseline visual acuity better than 20/200 whose visual acuity decreased to 20/200 or worse, confirmed at the next follow-up examination, were 17% by 1 year, 33% by 2 years, and 43% by 3 years after treatment. As soon as a poor vision outcome was observed, improvement of visual acuity to a level that no longer met the definition for a poor vision outcome was rare. Greater baseline tumor apical height and shorter distance between the tumor and the foveal avascular zone (FAZ) were the factors most strongly associated with loss of six or more lines of visual acuity after treatment. These two factors and baseline visual acuity also were strongly associated with visual acuity 20/200 or worse after treatment. Patient history of diabetes, presence of tumor-associated retinal detachment, and tumors that were not dome shaped also were associated with greater risk for both of the poor vision outcomes.

Conclusions: Forty-three percent to 49% of treated eyes had substantial impairment in visual acuity by 3 years after I^{125} brachytherapy, defined as a loss of six or more lines of visual acuity from the pretreatment level (49% of eyes) or visual acuity of 20/200 or worse (43% of eyes) that was confirmed at the next 6-month examination. Patients with a history of diabetes and patients whose eyes had thicker tumors, tumors close to or beneath the FAZ, tumor-associated retinal detachment, or tumors that were not dome shaped were those most likely to have a poor visual acuity outcome within 3 years after I^{125} brachytherapy. *Ophthalmology* 2001;108:348-366 © 2001 by the American Academy of Ophthalmology.

Choroidal melanoma is the most common primary malignant intraocular tumor in adults. Until the 1980s, the standard treatment of choroidal melanoma was enucleation.¹ However, alternative therapies that attempted to preserve the eye and vision were proposed as early as the 1930s. Diathermy for destruction of choroidal tumors was initiated in the mid 1930s.² Moore,³ in 1930, treated a patient's only remaining eye by implanting radon seeds directly into the tumor. Stallard⁴ also used radon seeds and later introduced cobalt 60 (C⁶⁰) episcleral plaques. By the mid 1980s, preliminary data based on limited case series published by Shields et al,⁵ Augsburger et al,⁶ and Adams et al⁷ on use of C⁶⁰ plaques, Packer et al⁸ on use of iodine 125 (I¹²⁵) plaques, Char and Castro⁹ on use of helium ion therapy, and Gragoudas et al¹⁰ on proton beam therapy, demonstrated that radiation techniques held promise for preservation of the eye and at least partial vision without decreasing life-span, for some patients.

The Collaborative Ocular Melanoma Study (COMS) is a set of multicenter clinical trials initiated in 1986 to address the role of radiotherapy versus enucleation with respect to long-term survival in selected patients with choroidal melanoma. I¹²⁵ brachytherapy was selected as the method of radiation delivery for patients with choroidal melanoma large enough to warrant treatment but not so large that enucleation was judged to be mandated. A review of factors considered in selecting I¹²⁵ has been published.^{11,12} Since the initiation of the COMS, a number of publications have appeared that report short-term visual acuity results in small case series and clinical trials of patients with choroidal melanoma treated with I¹²⁵ brachytherapy.^{13–25} Most studies were based on retrospective review of patient records; a protocol for visual acuity measurement typically was not described. Few investigators identified baseline and treatment factors that were associated with visual acuity outcome after I¹²⁵ brachytherapy. This report describes visual acuity outcomes and baseline and treatment factors associated with maintenance or loss of visual acuity during the first 3 years after treatment in the largest study of I¹²⁵ brachytherapy for choroidal melanoma conducted to date in which patients were prospectively enrolled according to well-defined inclusion and exclusion criteria, observed regularly with minimal loss to follow-up and in which visual acuity was measured following a standard protocol.

Methods

Patient Selection and Follow-up

The COMS design has been described previously.²⁶ An independent data and safety monitoring committee and the institutional review boards of all participating institutions reviewed and approved the design and protocol before patient enrollment began. Each enrolled patient gave written consent to random treatment assignment before enrollment. Patients eligible for the COMS trial of I¹²⁵ brachytherapy versus enucleation had unilateral choroidal melanoma at least 2.5 mm in height, but no more than 10.0 mm in height (no more than 8.0 mm in height whenever the tumor was near the disc), and no more than 16.0 mm in diameter. In addition to the size criteria, eligible patients were 21 years of age or older, had visual acuity in the fellow eye of better than 20/200, and were judged by the local COMS physicians to be suitable candidates for radiotherapy and surgery. Patients were ineligible for the COMS if they had been diagnosed with any other malignancy (excluding superficial nonmelanotic skin cancer or cancer in situ of the uterine cervix), currently were receiving immunosuppressive therapy, were unable to return for follow-up examinations, or had any other significant health problem that threatened survival for at least 5 years. Patients whose tumor could not be clearly delineated by echograms, could not be observed ophthalmoscopically because of insufficient clarity of the ocular media, was diffuse or multifocal, was contiguous with the optic disc or within 2 mm of the optic disc and the ophthalmologist judged that it could not be plaqued satisfactorily, was predominately a ciliary body tumor, involved the angle or iris, had detectable extrascleral extension of more than 2 mm, had a fine-needle aspiration biopsy, or had been treated previously also were ineligible. Diagnosis of neovascular or secondary glaucoma in the tumor-containing eye or treatment for a condition secondary to the melanoma also rendered the patient ineligible.

Detailed information from a patient interview and ophthalmic evaluation was collected on standard forms at the baseline examination, every 6 months thereafter for 5 years, and subsequently once yearly.²⁷ The ophthalmic examination included best-corrected visual acuity measurement of each eye, ophthalmoscopic examination of both eyes, measurement of intraocular pressure, and gonioscopy when indicated. Visual acuity was measured by a COMS-certified examiner according to a standard protocol that used best refractive correction and a Bailey-Lovie chart.²⁸ No attempt was made to mask visual acuity examiners to the identity of the treated eye; in some cases, the examiner was aware of which eye had been treated, in other cases not. Standardized A-scan and contact B-scan echography was performed by a COMS-certified echographer at baseline for all patients and at follow-up examinations for all patients treated with I¹²⁵ brachytherapy. Tumor shape was assessed from baseline photoechograms sent to the COMS Echography Center. Patients were assigned randomly to I¹²⁵ brachytherapy or enucleation by the COMS Coordinating Center.

The first patient enrolled in the trial of I¹²⁵ brachytherapy in February 1987. At completion of accrual on July 31, 1998, 1317 patients had enrolled. Of these, 657 patients were randomly assigned to I¹²⁵ brachytherapy. Seven patients (1.1%) who declined their assigned treatment were excluded from this report, as were an additional 27 patients (4.1%) who had no or only one follow-up visual acuity examination. Most patients in the latter subgroup died before 1 year of follow-up or were being observed by their local physicians. The remaining 623 patients are the subject of this report.

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Treatment Protocol

The size of the plaque used for I^{125} brachytherapy was chosen so that the tumor base, with a tumor-free margin of 2 to 3 mm on all borders, was covered entirely by the plaque. An exception was made for tumors within 2 mm of the optic nerve; a notched or trimmed plaque with its edge placed between the margin of the tumor and the optic nerve could be used for these cases. For tumors 5.0 mm or more in apical height, the radiation dose was prescribed to the apex of the tumor. For tumors 2.5 to 4.9 mm in apical height, the prescription point was 5 mm from the interior surface of the sclera. In 1996, the COMS adopted the recommendations of the American Association of Physics in Medicine Task Group 43 (TG-43) for dosimetry of interstitial sources^{29,30}; all reported doses have been converted to this scale. Under the TG-43 formalism, the absorbed dose at the prescription point was 85 Gy delivered at the rate of at least 0.42 Gy/hr, but no more than 1.05 Gy/hr. The COMS protocol specified that treatment take place within 28 days of COMS enrollment; 587 patients (94.2%) included in this analysis met this protocol specification.

Statistical Methods

This analysis is based on data received at the COMS Coordinating Center by August 31, 1999. As of this date, all patients had been enrolled in the COMS for at least 1 year, and most had been observed for at least 3 years, allowing reliable estimation of effects of baseline and treatment characteristics on visual acuity within this period. Information for later follow-up times currently is less complete. Visual acuity measurements performed by an individual other than a COMS-certified examiner were included in analyses whenever a COMS-approved chart was used and the test conditions and results conformed to COMS protocol. A visual acuity measurement was not available for 6.4% of expected patient examinations, usually because the patient missed the examination. For analysis using the life-table and Cox proportional hazards models, missing visual acuity measurements were interpolated as the average of the measurements at the previous and following examinations when these were available; otherwise, the data were censored after the last actual follow-up examination. A total of 3.6% of expected measurements were interpolated. Interpolated measurements were not included in reported distributions of visual acuity. Patients who died during the 3-year follow-up period and who had not experienced a poor vision outcome were considered censored after their last follow-up examination.

Two poor vision outcomes, a visual acuity of 20/200 or worse that was confirmed at the next follow-up examination, and loss of six lines or more of visual acuity from baseline, that is, a quadrupling of the minimum angle of resolution, that was confirmed at the next follow-up examination, were the focus of analyses to identify baseline and treatment factors associated with poor visual acuity outcome during follow-up. Patients whose eyes were enucleated before their 3-year follow-up examination were counted as having a poor vision outcome in all analyses. Many eyes that were enucleated had met the definition for a poor vision outcome before the enucleation and were counted as having a poor vision outcome at that time; the remainder were counted as having a poor vision outcome at the study examination after the enucleation. For computation of mean visual acuity during follow-up, enucleated eyes were counted as having visual acuity of 20/2500, that is, two lines less than 20/1600, the smallest measurable visual acuity in the COMS. Each of the baseline characteristics was tested individually for association with each of the two poor vision outcomes using life-table analysis with the log rank test.³¹ Intervals for the life-table analysis were chosen to correspond to the COMS examination schedule during the first 5 years after enrollment, which

specified a follow-up examination every 6 months with an allowable 6 month window centered on enrollment anniversary dates. Characteristics that did not have at least 15 patients within each subgroup of the characteristic were not tested for association with the outcome. Because the life-table model does not allow for recovery from a poor vision outcome, a two-state stochastic Markov model was used to estimate prevalence of the outcomes at each follow-up time³²; these prevalences were adjusted for patients who had a poor vision outcome, but later recover vision.

Characteristics showing univariate association with a poor vision outcome at a statistical significance level of $P < 0.20$ by the log rank test were included in a multivariate analysis using the Cox proportional hazards model.³³ Twenty-five patients, who were not at risk of losing six or more lines because their baseline visual acuity was within six lines of the smallest measurable level under COMS protocol, were excluded from the multivariate analysis of that outcome, as were the five patients who did not have a baseline visual acuity measurement. The 556 patients with baseline visual acuity better than 20/200 were included in the multivariate analysis of time to visual acuity 20/200 or worse at the first of two consecutive examinations. A backwards stepwise procedure that included patients with complete information and retained characteristics that were significantly associated with the outcome at a statistical significance level of $P \leq 0.05$ was used to select the best multivariate model. Once the "best" model was selected, patients with missing information for characteristics not in the best model were added to the analysis to create a final model.

Results

Description of Patient Population and Treatment

Demographic, tumor, and treatment characteristics are described in Table 1. Fifty-one percent of the patients were female; ages at enrollment ranged from 24 to 88 years, with a median of 61 years. Ninety-eight percent of the patients were non-Hispanic white.

Tumor apical height ranged from 2.5 to 10.0 mm, consistent with eligibility criteria, with a median of 4.2 mm. Largest basal dimension ranged from 4.5 to 16.0 mm, with a median of 11.5 mm. More than half of the tumors (54%) were located entirely posterior to the equator; 15% were 2.0 mm or closer to the optic disc, and 13% were beneath the foveal avascular zone (FAZ). Twenty-two percent of patients had a tumor-associated retinal detachment that extended into the FAZ while the patient was in the upright or supine position. Most tumors (77%) were dome shaped; a minority (16%) had a collar-button configuration. There were small numbers (<7%) of tumors with other shapes, including irregular, flat, peaked, and lobulated.

At baseline, 70% of eyes with choroidal melanoma (study eyes) had visual acuity of 20/40 or better; 10% had 20/200 or worse visual acuity. Only 3% of fellow eyes had visual acuity worse than 20/40; patients with visual acuity of 20/200 or worse in the fellow eye were not eligible for this COMS clinical trial. Approximately 20% of patients had a history of a vision-limiting or threatening condition other than choroidal melanoma; these included cataract, age-related macular degeneration, diabetic retinopathy, glaucoma, and amblyopia. Other than cataract, fewer than 2% of study eyes had a history of these conditions or any other vision-limiting or threatening condition.

Distribution of radiation doses received by the foveola, optic disc, and lens are described in Table 1. The doses to the center of each ocular structure were calculated based on reported plaque location and I^{125} seed activity and location in the plaque, and do not account for anisotropy or shielding by the plaque lip. Hence, reported doses represent a potential maximum dose of radiation

Table 1. Baseline Demographic, Ophthalmic, Tumor, and Treatment Characteristics of Study Patients

Characteristic	n	%
Demographic characteristics		
Gender		
Female	315	50.6
Male	308	49.4
Race		
White	611	98.1
Other	12	1.9
Age (yrs)		
<50	160	25.7
50–69	318	51.0
≥70	145	23.3
Median age	61	
General medical characteristics		
History of diabetes		
No	562	90.2
Yes	59	9.5
Unknown	2	0.3
History of hypertension		
No	451	72.4
Yes	171	27.4
Unknown	1	0.2
Ophthalmic characteristics		
Visual acuity in study eye		
≥20/20	208	33.4
20/25–20/40	230	36.9
20/50–20/160	118	18.9
≤20/200	61	9.8
Not available	5	1.0
Median visual acuity	20/32	
Visual acuity in fellow eye		
≥20/20	446	71.6
20/25–20/40	156	25.0
20/50–20/160	18	2.9
Not available	3	0.5
Median visual acuity	20/20	
Lens status in study eye		
Phakic	587	94.2
Aphakic or pseudophakic	36	5.8
History of diabetic retinopathy		
No	612	98.2
Yes	11	1.8
History of any vision-limiting or threatening condition:		
Study eye		
No	501	80.4
Yes	122	19.6
Fellow eye		
No	511	82.0
Yes	112	18.0
Intraocular pressure (mmHg)		
Study eye		
≤10.0	128	20.5
10.1–15.0	233	37.4
15.1–20.0	188	30.2
>20.0	71	11.4
Not available	3	0.5
Fellow eye		
≤10.0	88	14.1
10.1–15.0	220	35.3
15.1–20.0	214	34.3
>20.0	98	15.7
Not available	3	0.5

(continues)

Table 1. (continued)

Characteristic	n	%
Tumor characteristics		
Apical height (mm)		
2.5–5.0	403	64.7
5.1–7.5	157	25.2
7.6–10.0	63	10.1
Median height	4.2	
Largest basal dimension (mm)		
4.5–8.0	84	13.5
8.1–11.0	214	34.3
11.1–14.0	229	36.8
14.1–16.0	96	15.4
Median diameter	11.5	
Location of anterior tumor border		
Pars plicata	22	3.5
Pars plana	49	7.9
Equator to ora serrata	214	34.3
Posterior to equator	338	54.3
Location of posterior tumor border		
Equator to ora serrata	15	2.4
Posterior to equator	608	97.6
Ciliary body involvement		
No	565	90.7
Yes	53	8.5
Unavailable	5	0.8
Tumor-associated retinal detachment		
Over foveal avascular zone while patient in upright or supine position	139	22.3
Not involving foveal avascular zone	210	33.7
No retinal detachment	274	44.0
Distance from closest border of tumor to center of foveal avascular zone (mm)		
0.0	82	13.2
0.1–2.0	155	24.9
2.1–5.0	183	29.4
5.1–8.0	102	16.4
>8.0	95	15.2
Not available	6	1.0
Median distance	3.0	
Distance from closest border of tumor to edge of optic disc (mm)		
0.0–2.0	93	14.9
2.1–4.0	194	31.1
4.1–6.0	151	24.2
6.1–8.0	92	14.8
>8.0	91	14.6
Not available	2	0.3
Median distance	4.5	
Tumor shape		
Dome	480	77.0
Collar button	101	16.2
Other	42	6.7
Treatment characteristics		
Muscle detachment performed		
No	247	39.6
Yes	376	60.4
Prescription point		
Apex of tumor	236	37.9
5 mm from interior surface of sclera	387	62.1
Dose to prescription point (Gy)		
≤80.0	19	3.0
80.1–85.0	344	55.2
85.1–90.0	243	39.0
>90.0	17	2.7
Median dose	85.0	

(continues)

Table 1. (continued)

Characteristic	n	%
Dose rate at prescription point (cGy/hr)		
<55.0	169	27.1
55.0–69.9	134	21.5
70.0–84.9	153	24.6
≥85.0	167	26.8
Median dose rate	70.5	
Radiation dose to foveola (Gy)		
<40.0	165	26.5
40.0–79.9	149	23.9
80.0–149.9	146	23.4
≥150.0	159	25.5
Not available	4	0.6
Median dose	79.0	
Radiation dose to center of optic nerve head (Gy)		
<30.0	135	21.7
30.0–49.9	161	25.8
50.0–74.9	162	26.0
≥75.0	159	25.5
Not available	6	1.0
Median dose	52.1	
Radiation dose to center of lens (Gy)		
<12.0	147	23.6
12.0–15.9	168	27.0
16.0–23.9	167	26.8
≥24.0	136	21.8
Not available	5	0.8
Median dose	15.6	
Radiation dose to retina opposite plaque center (Gy)		
≤7.0	158	25.4
7.1–8.0	166	26.6
8.1–9.0	104	16.7
>9.0	195	31.3
Median dose	7.9	
Plaque type and size		
Standard 12 mm	69	11.1
Standard 14 mm	128	20.5
Standard 16 mm	158	25.4
Standard 18 mm	126	20.2
Standard 20 mm	101	16.2
Customized or notched	41	6.6

received by the structure. Dose to the foveola ranged from 6 to 433 Gy, with a median of 79 Gy and an interquartile range (25th to 75th percentile) of 39 to 151 Gy. Dose to the optic disc ranged

from 5 to 275 Gy, with a median of 52 Gy and an interquartile range of 32 to 75 Gy. Dose to the lens ranged from 8 to 98 Gy, with a median of 16 Gy and an interquartile range of 12 to 23 Gy. Calculations of doses to critical structures could not be performed for a few patients (<1%).

Visual Acuity

All 623 patients included in this analysis had been enrolled in the COMS for at least 1 year and were eligible for 6-month and 1-year follow-up examinations. Ninety-three percent and 80% of patients had been enrolled for at least 2 and 3 years, respectively, and were eligible for these examinations. Percentages of patients eligible for each follow-up examination who had visual acuity measurements taken ranged from 98% at 6 months to 86% at 42 months.

Distributions of visual acuities by year of follow-up appear in Table 2. Median visual acuity in the study eye was 20/32 at baseline and decreased to 20/125 by the third year of follow-up. The study eye of 40 patients included in these analyses was enucleated before the 3-year follow-up examination. The number reported as enucleated in Table 2 reflects the number of patients who had undergone enucleation and had completed the requisite follow-up examination; not all patients whose eyes were enucleated were eligible for a 3-year follow-up examination as of the cutoff date for this report, nor did all patients eligible for 3-year follow-up complete all required follow-up examinations.

Loss of Six or More Lines of Visual Acuity from Baseline. Life-table estimates of percentage of patients who lost six or more lines of visual acuity from baseline, as confirmed at the next follow-up examination, were 18% by 1 year (95% confidence interval [CI], 15%–21%), 34% by 2 years (95% CI, 30%–38%), and 49% by 3 years of follow-up (95% CI, 44%–53%; Table 3). These percentages include patients whose eyes were enucleated before the examination in question. Recovery from a loss of six or more lines after treatment was rare; only 10 of 244 patients with this outcome (4.1%) ever recovered visual acuity to within six lines of the pretreatment level at any subsequent examination during the 3-year follow-up period. A stochastic Markov model that allowed for modeling both vision loss and recovery, with loss defined as a decrease from baseline of six or more lines confirmed at the next examination, and recovery defined as visual acuity within six lines of baseline at any examination after a loss, gave nearly the same rates as the life-table model: 18% versus 17% at 1 year, 34% versus 32% at 2 years, and 49% versus 46% at 3 years for the life-table and Markov models, respectively.

Gender was the only demographic characteristic significantly associated with time to loss of six or more lines at two consecutive

Table 2. Distribution of Visual Acuity in the Plaque Eye by Months since Enrollment

Visual Acuity	Months since Enrollment							
	0		12		24		36	
	n	%	n	%	n	%	n	%
≥20/20	208	33.4	152	24.4	97	16.8	62	12.4
20/25–20/40	230	36.9	203	32.6	136	23.6	93	18.6
20/50–20/80	80	12.8	70	11.2	75	13.0	53	10.6
20/100–20/160	38	6.1	35	5.6	34	5.9	43	8.6
20/200–20/320	25	4.0	37	5.9	39	6.8	32	6.4
20/400–20/640	13	2.1	27	4.3	27	4.7	24	4.8
≤20/800	23	3.7	73	11.7	98	17.0	116	23.2
Enucleated	0	0.0	9	1.4	23	4.0	31	6.2
Not available	6	1.0	17	2.7	48	8.3	47	9.4
Total Patients	623	100	623	100	577	100	501	100
Median visual acuity	20/32		20/40		20/50		20/125	

Table 3. Life-table Estimates of Time to Visual Acuity Outcomes

Visual Acuity Outcome	Midpoint of Interval (mos)	No. at Risk at Start of Interval	No. of Events during Interval	No. Censored during Interval	Cumulative Percentage with Event	95% Confidence Interval
Loss of six or more lines, with confirmation at the next visit	6	589	60	27	10	(8–13)
	12	502	40	29	18	(15–21)
	18	433	35	29	25	(21–28)
	24	369	45	21	34	(30–38)
	30	303	33	19	42	(37–46)
	36	251	31	16	49	(44–53)
Visual acuity 20/200 or less, with confirmation at the next visit	6	556	56	25	10	(8–13)
	12	475	33	26	17	(14–20)
	18	416	37	27	24	(21–28)
	24	352	38	21	33	(29–37)
	30	293	17	24	37	(33–41)
	36	252	24	15	43	(38–48)

follow-up examinations on univariate testing, with males having higher risk (Table 4). Race was not tested for association with this outcome because there were small numbers of minorities (Table 1). Medical and ophthalmic characteristics that were associated with higher risk of this outcome on univariate analysis included poor baseline visual acuity in the study eye and history of diabetes. Visual acuity in the fellow eye, aphakia or pseudophakia in the eye with melanoma, a history of hypertension, intraocular pressure in either eye, and history of a vision-threatening or vision-limiting condition in either eye were not associated with this outcome.

Tumor characteristics associated with increased risk of this outcome included greater tumor thickness, greater tumor basal dimension (>14.0 mm), more posterior tumor location (as indexed by location of the anterior tumor border, distance from the tumor to the FAZ, and distance from the tumor to the optic disc), presence of tumor-associated retinal detachment, and tumor shape other than dome. With respect to treatment factors, higher radiation doses to the foveola, center of the optic disc, and retina opposite the plaque center were significantly associated with this outcome, but dose to the lens, dose to the prescription point, and dose rate were not. As expected based on the treatment protocol, the range of doses to the prescription point was narrow; hence, this study had little power to detect effects associated with variation in this treatment parameter. Plaque size was marginally associated with risk ($P = 0.048$), with smaller plaque sizes having lower risk. Muscle detachment during plaque surgery was not associated with this visual acuity outcome.

Variables that were associated with time to loss of six lines or more in multivariate analysis are listed in Table 5. Characteristics that were associated with increased risk of the outcome included a history of diabetes, greater tumor apical height, tumor shape other than dome, tumor 2.0 mm or closer to the FAZ, and presence of tumor-associated retinal detachment, particularly when extending into the FAZ.

Decrease in Visual Acuity to 20/200 or Worse. For patients with baseline visual acuity better than 20/200, life-table estimates of time to best-corrected visual acuity of 20/200 or worse in the plaqued eye, confirmed at the next follow-up examination, are presented in Table 3. Seventeen percent of these patients had experienced visual acuity 20/200 or worse by 1 year of follow-up and had this finding confirmed at their next examination; by 3 years, this figure was 43% (95% CI, 38%–48%). As for the previous analysis, patients who experienced this outcome rarely recovered visual acuity better than 20/200; of the 187 patients who lost visual acuity to 20/200 or worse, nine patients recovered visual acuity better than 20/200 at any subsequent examination, but three

of these patients did not maintain recovered visual acuity during the remainder of the 3-year follow-up period.

Baseline and treatment factors associated with follow-up visual acuity of 20/200 or worse were similar to the factors associated with a 6-line or more loss of visual acuity (Table 5) and included tumor apical height, distance from the tumor to the FAZ, tumor shape, and diabetes. Baseline visual acuity was strongly associated with time to visual acuity of 20/200 or worse; patients with better baseline visual acuity (20/40 or better) were less likely to lose visual acuity to the 20/200 or worse level within 3 years of treatment as compared with patients with worse baseline visual acuity (20/50–20/160). Figure 1 shows mean visual acuity over follow-up stratified by the factors associated with both poor vision outcomes; mean visual acuity by the number of these factors, or “high risk characteristics,” is shown in Figure 2.

Other Visual Acuity Outcomes. Radiation treatment sometimes was followed by improvement in visual acuity. Of the 61 plaqued eyes with baseline visual acuity of 20/200 or worse, 28% recovered visual acuity better than 20/200, and 18% better than 20/100 by the 6-month follow-up examination. An additional 10% of plaqued eyes recovered visual acuity better than 20/200 at one or more examinations between the 6- and 24-month follow-up examinations. Only one of the 61 eyes recovered visual acuity better than 20/200 after the 24-month follow-up examination. Half of the eyes that recovered vision maintained visual acuity better than 20/200 for 18 months or more; however, most did not maintain recovered vision over the remainder of follow-up. At 3 years, the two-state Markov estimate of proportion of eyes in this small subgroup with visual acuity better than 20/200 was 10%.

In the 118 eyes with baseline visual acuity between 20/50 and 20/160, 20% recovered visual acuity better than 20/50 by the 6-month follow-up examination, and 35% had at least one visual acuity measurement better than 20/50 by the 3-year follow-up examination. However, the two-state Markov model estimate of the proportion of eyes in this subgroup with visual acuity better than 20/50 after 3 years of follow-up was only 10%, indicating that most of these improvements in visual acuity were not sustained. Similarly, 20% of the 179 eyes with baseline visual acuity of 20/50 or less gained three or more lines of visual acuity by 6 months after treatment; the Markov estimate of percentage of patients in this group with visual acuity three lines or more above baseline at 3 years was only 7%.

Although specific reasons for visual acuity improvement were not collected, available data were examined to determine possible reasons for improvement. Recovery between baseline and the 6-month follow-up examination appeared most often to be the

Table 4. Cumulative Percentages of Patients with Visual Acuity Outcome in the Treated Eye at 3 Years Stratified by Baseline and Treatment Characteristics

Characteristic	Visual Acuity Outcome			
	Lost ≥ 6 Lines		Visual Acuity $\leq 20/200$	
	No. with Event	% with Event*	No. with Event	% with Event*
<i>Demographic characteristics</i>				
Gender				
Female	110	44.6	92	39.2
Male	134	53.4	113	46.9
<i>General medical characteristics</i>				
History of diabetes				
No	210	46.5	175	40.6
Yes	32	71.7	28	64.3
<i>Ophthalmic characteristics</i>				
Visual acuity in eye with tumor				
$\geq 20/20$	63	36.6	43	25.6
20/25–20/40	98	49.8	85	43.2
20/50–20/160	69	68.1	77	73.1
$\leq 20/200$	14	49.6	—	—
<i>Tumor characteristics</i>				
Apical height (mm)				
2.5–5.0	128	39.5	90	26.9
5.1–7.5	77	62.9	65	47.7
7.6–10.0	39	79.3	32	76.1
Largest basal dimension (mm)				
4.5–8.0	32	44.4	25	35.8
8.1–11.0	83	51.7	66	41.3
11.1–14.0	82	46.3	71	41.6
14.1–16.0	47	61.3	43	57.4
Location of anterior tumor border				
Pars plicata or pars plana	16	28.6	15	26.9
Equator to ora serrata	76	43.5	70	40.9
Posterior to equator	152	56.4	120	48.1
Tumor-associated retinal detachment				
Over foveal avascular zone while patient in upright or supine position	73	69.4	61	67.3
Not involving foveal avascular zone	94	55.1	82	49.7
No detachment	77	35.0	62	28.4
Distance tumor to foveal avascular zone (mm)				
0.0	44	70.2	38	73.9
0.1–2.0	75	64.5	61	54.7
2.1–5.0	68	42.2	58	37.0
5.1–8.0	34	41.9	30	36.7
>8.0	21	28.5	17	24.4
Distance tumor to disc (mm)				
0.0–2.0	47	64.1	39	59.8
2.1–4.0	85	59.9	66	49.8
4.1–6.0	62	47.5	59	46.5
6.1–8.0	30	39.2	26	34.6
>8.0	20	26.5	15	19.5
Tumor shape				
Dome	172	44.4	137	36.9
Collar button or other	72	66.2	68	66.1
<i>Treatment characteristics</i>				
Radiation dose to foveola (Gy)				
<40.0	41	31.2	35	26.5
40.0–79.9	56	46.4	48	40.6
80.0–149.9	61	50.2	52	44.6
≥ 150.0	86	70.6	70	65.9
Radiation dose to optic disc (Gy)				
<30.0	28	25.3	24	20.8
30.0–49.9	65	50.1	55	43.1
50.0–74.9	66	50.0	58	46.9
≥ 75.0	85	69.3	68	62.9
Radiation dose to retina opposite the plaque center (Gy)				
≤ 7.0	55	41.9	39	31.2
7.1–8.0	58	43.5	48	37.1
8.1–9.0	27	33.3	24	30.8
>9.0	104	69.1	94	65.6
Plaque type and size (mm)				
Standard 12 or 14	79	48.1	60	39.2
Standard 16 or 18	99	45.9	86	40.3
Standard 20	47	56.4	44	53.5
Customized or notched	19	61.8	15	55.6 [†]

* Life-table estimate of cumulative percentage with the outcome at 3 years.

[†] $P = 0.052$ for this outcome.

Table 5. Multivariate Cox Proportional Hazards Models for Association of Baseline and Treatment Variables with Visual Acuity Outcomes

Visual Outcome	Variable	Relative Risk	95% Confidence Interval	Chi-square Result	Degrees of Freedom	P Value
Loss of 6 or more lines, with confirmation, at the next examination	<i>Medical history</i>					
	History of diabetes			17.9	1	<0.0001
	No (ref.)	1.00				
	Yes	2.75	1.78–4.25			
	<i>Tumor characteristics</i>					
	Apical height (mm)			39.4	2	<0.0001
	2.5–5.0 (ref.)	1.00				
	5.1–7.5	1.90	1.36–2.65			
	7.6–10.0	4.72	2.90–7.66			
	Tumor shape			14.1	1	0.0002
	Dome (ref.)	1.00				
	Collar button or other	2.00	1.41–2.85			
	Distance from tumor to foveal avascular zone (mm)			48.5	4	<0.0001
	0.0	5.90	3.22–10.8			
	0.1–2.0	3.99	2.29–6.97			
	2.1–5.0	1.95	1.15–3.33			
	5.1–8.0	1.86	1.03–3.36			
	>8.0 (ref.)	1.00				
	Tumor associated retinal detachment			9.8	2	0.0075
Visual acuity 20/200 or less, with confirmation at the next examination	<i>Medical history</i>					
	History of diabetes			16.4	1	<0.0001
	No (ref.)	1.00				
	Yes	2.94	1.80–4.81			
	<i>Ophthalmic characteristics</i>					
	Baseline visual acuity			30.4	2	<0.0001
	≥20/20 (ref.)	1.00				
	20/25–20/40	1.57	1.04–2.37			
	20/50–20/160	3.60	2.25–5.75			
	<i>Tumor characteristics</i>					
	Apical height (mm)			37.5	2	<0.0001
	2.5–5.0 (ref.)	1.00				
	5.1–7.5	1.90	1.30–2.77			
	7.6–10.0	5.34	3.14–9.07			
	Tumor shape			15.7	1	<0.0001
	Dome (ref.)	1.00				
	Collar button or other	2.23	1.53–3.29			
	Distance from tumor to foveal avascular zone (mm)			41.3	4	<0.0001
	0.0	7.25	3.57–14.7			
	0.1–2.0	4.24	2.26–7.97			
	2.1–5.0	2.17	1.19–3.95			
	5.1–8.0	2.41	1.25–4.66			
	>8.0 (ref.)	1.00				
	Tumor associated retinal detachment			11.9	2	0.0026
	Over the foveal avascular zone	1.89	1.24–2.90			
	Elsewhere	1.78	1.21–2.63			
	No detachment (ref.)	1.00				

(ref.) = reference group.

result of resolution of tumor-associated retinal detachment. Recoveries later in follow-up occasionally were observed after cataract extraction, clearing of vitreous hemorrhage, or apparent patient adaptation to using peripheral vision after loss of central vision. However, for a high proportion of cases with improved visual acuity during follow-up (25%), there was no reason for improvement that could be deduced from the data reported. In an equal proportion of cases with improvement during follow-up, interpretation of visual acuity changes was confounded by the presence of another vision-threatening condition.

Thus, although some patients experienced periods of improve-

ment after treatment, improvement often was transient. At 3 years of follow-up, approximately half (57%) of patients in the group as a whole had visual acuity that was three or more lines worse than the pretreatment level or the study eye had been enucleated, whereas 3% had visual acuity that was three or more lines better than at baseline.

Most patients had good visual acuity in the fellow eye at baseline and throughout the follow-up period. Patients with 20/200 or worse visual acuity in the fellow eye at baseline were ineligible for the study; at 3 years of follow-up, 2 of 456 patients (0.4%) had 20/200 or worse visual acuity in the fellow eye, and 17 patients

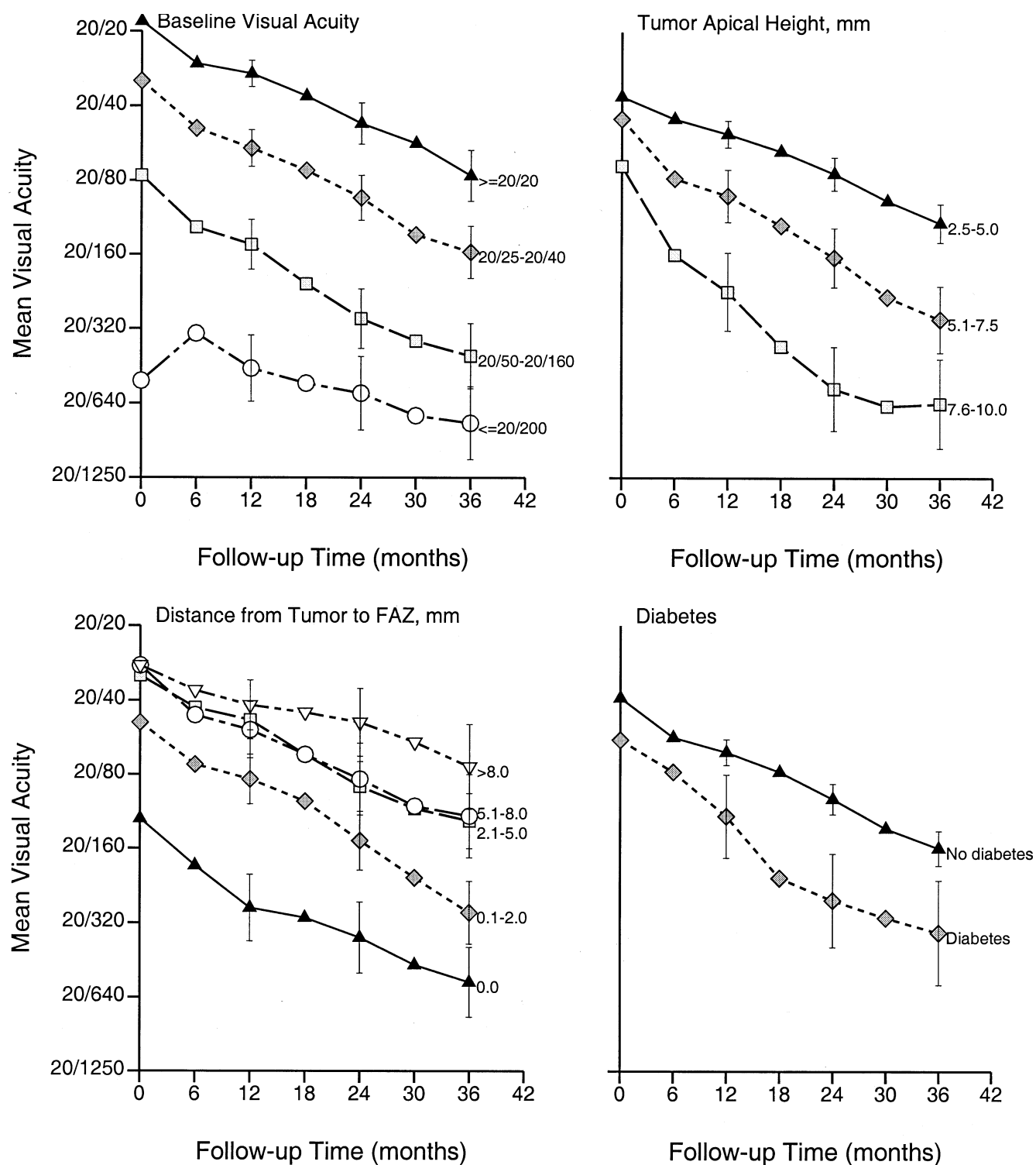


Figure 1. Mean visual acuity with 95% confidence intervals by baseline risk factors; 95% confidence intervals for means are indicated by the error bars. FAZ = foveal avascular zone.

(3.7%) had visual acuity of 20/50 to 20/160 in the fellow eye. Approximately 1% to 2% of patients had 20/40 or better visual acuity in the study eye and visual acuity worse than 20/40 in the fellow eye at any given follow-up examination.

In summary, visual acuity during the first 3 years after ^{125}I plaque radiotherapy for choroidal melanoma declined at a rate of

approximately two lines per year on average, although this time course was highly variable from patient to patient. Forty-three to 49% of treated eyes had substantial impairment in visual acuity by 3 years after treatment, defined as a loss of six or more lines of visual acuity from the pretreatment level (49% of eyes) or a decrease in visual acuity from better than 20/200 at baseline to

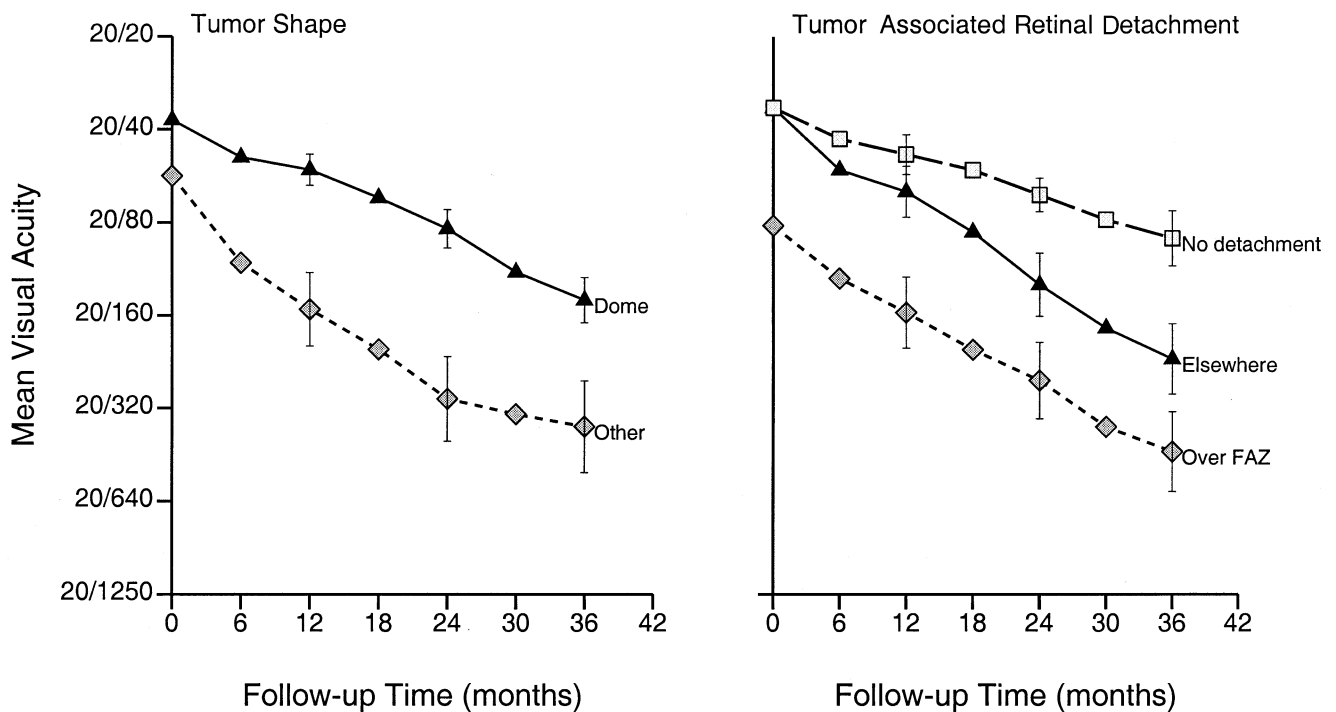


Figure 1 (continued). Mean visual acuity with 95% confidence intervals by baseline risk factors; 95% confidence intervals for means are indicated by the error bars. FAZ = foveal avascular zone.

20/200 or worse during follow-up (43% of eyes). Thirty-nine percent of eyes had visual acuity within three lines of baseline at the 3-year follow-up examination. By 3 years after treatment, 13% of eyes had visual acuity three to six lines less than baseline, whereas 3% had visual acuity three or more lines more than baseline. The large subgroup of patients with no high risk characteristics, that is, patients without diabetes with dome-shaped tumors 5.0 mm or less in apical height that were more than 2.0 mm away from the FAZ and did not have retinal detachment, maintained on average 20/40 or better visual acuity throughout the 3-year follow-up period. In these patients, the probability of losing six or more lines within 3 years was 12%, and the probability of 20/200 or worse visual acuity was 9%. However, 32% of eyes in this favorable subgroup had a three-line or more decrease in visual acuity that was confirmed at the next follow-up examination within the 3 years of follow-up.

Discussion

Most factors identified in this analysis of COMS data to be associated with visual acuity outcome are not surprising. Eyes with thicker tumors and tumors beneath the FAZ received higher overall doses of radiation to this critical structure. They also tended to have poorer baseline visual acuity because of direct involvement of the macula or tumor-associated retinal detachment involving the macula. Baseline visual acuity was very strongly related to follow-up level of visual acuity and less strongly related to loss of visual acuity because, on average, plaqued eyes with all but the poorest initial visual acuities lost visual acuity at similar rates, that is, approximately two lines per year after treatment (Fig 1). However, patients with one or more

high-risk characteristics tended to lose vision at a faster rate than did patients with no high-risk characteristics (Fig 2). Patients with medium-sized choroidal melanoma that meet COMS eligibility criteria would be expected to have similar visual acuity results if treated with I¹²⁵ brachytherapy according to COMS treatment parameters.

It is not clear why eyes with dome-shaped tumors had better visual acuity outcome than eyes with tumors of collar button and other shapes. These eyes were less likely to be enucleated during follow-up; however, within the subgroup of eyes that were not enucleated, eyes with dome-shaped tumors still had better visual acuity outcomes than eyes with tumors of collar button and other shapes. Tumors of collar button and other shapes also were significantly thicker on average than dome-shaped tumors (6.4 vs. 4.3 mm); however, when analyses were limited to relatively narrow ranges of tumor thickness, those with tumors of collar button and other shapes still were more likely to experience a poor vision outcome. Those with tumors of collar-button and other shapes also were more likely to have tumor-associated retinal detachment at baseline, less likely to have resolution of retinal detachment during follow-up, and more likely to experience new retinal detachment during follow-up when it had not been previously present; but again, an analysis that adjusted for presence or absence of retinal detachment at each follow-up examination, in addition to tumor apical height, did not affect the estimated relative risk associated with tumor shape. Hence, although nondome-shaped tumors are more likely to have high-risk characteristics, this does not appear to explain entirely the increased likelihood of experiencing a poor vision outcome associated

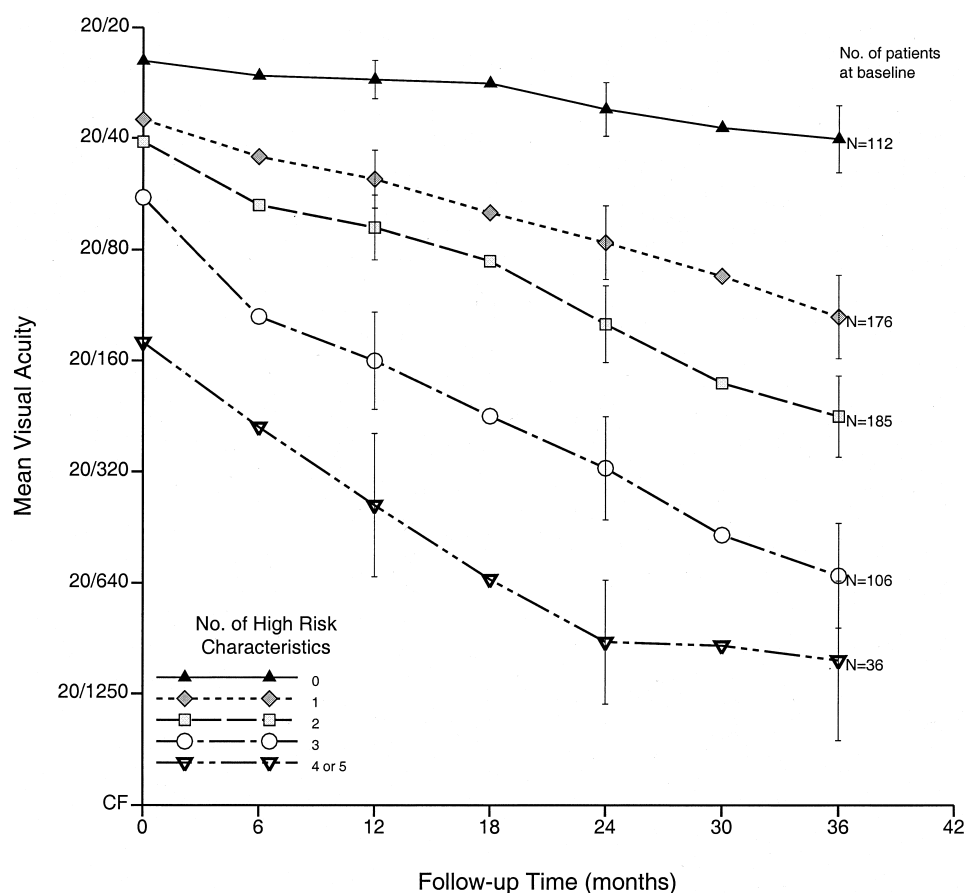


Figure 2. Mean visual acuity at examinations during foveal avascular zone the first 3 years of follow-up by number of high-risk characteristics. High risk characteristics were: tumor height ≥ 5.0 mm, distance between tumor and foveal avascular zone ≤ 2.0 mm, diabetes, nondome-shaped tumor, and presence of tumor-associated retinal detachment. Nondome-shaped tumors included tumors with collar button, flat, peaked, lobulated, and irregular shapes. 95% confidence intervals for the means are indicated by the error bars.

with these tumors. A recent study by Gündüz et al³⁴ that examined outcomes in predominantly ciliary body melanomas treated with brachytherapy using a variety of isotopes also reported that collar button shape was associated with greater likelihood of a three-line decrease in visual acuity during follow-up.

Direct comparison of COMS results with other published studies reporting visual acuity after I¹²⁵ brachytherapy for choroidal melanoma is problematic. Sizes of tumors treated vary across studies, as do radiation dose prescribed to the tumor and definitions of visual acuity outcomes; visual acuity results are highly dependent on these factors. Also, COMS visual acuity measurements were made with best correction, as determined by a standard refraction protocol, and followed a standard measurement protocol that included Bailey-Lovie charts and standard lighting; they may not be comparable with visual acuity measurements obtained during a routine clinical examination. Furthermore, many studies report only summary statistics based on visual acuity at "last follow-up" for patients with varying lengths of follow-up; because there is a strong relationship between visual acuity and time since treatment, these summary statistics are dependent on the exact distribution of follow-up times in the study and are not comparable across studies.

In a phase I clinical trial, Char et al¹⁹ found that, among 147 patients treated with I¹²⁵ plaque, 48% of eyes with initial visual acuity of 20/40 or better maintained 20/40 or better visual acuity after 12 months; in the COMS, this

figure was 68%. The mean apical height of tumors in the COMS was smaller than in Char et al's study, 4.3 versus 5.5 mm, which may account for this difference; however, tumors in Char et al's study received a lower dose of radiation to the prescription point than tumors in the COMS. A more recent study by Char et al²⁰ reported 2- to 3-year follow-up visual acuities in 73 patients enrolled in a phase III clinical trial and treated with I¹²⁵ brachytherapy: 26% were 20/15 to 20/40, 22% were 20/50 to 20/400, and 52% were worse than 20/400, with enucleated eyes included in the latter group. Distribution of initial visual acuity was nearly identical to that in the COMS. At the 30-month COMS follow-up examination, 39% were 20/20 to 20/40, 28% were 20/50 to 20/400, and 33% were worse than 20/400, including enucleations. Again, tumors in Char et al's study were larger on average and received a smaller dose of radiation than those in the COMS. Packer et al¹⁴ reported that 62% of 64 patients treated with 80 Gy to the tumor apex (not TG-43 standard) had experienced an adverse visual event in the treated eye by 3 years, where an adverse visual event was defined as visual acuity 20/100 or worse, or two or more lines of visual acuity lost. In the COMS, this figure was 78%, regardless of whether eyes with baseline visual acuity of 20/100 or worse are included. Packer et al's study included patients with tumors larger than in the COMS, but did not report pretreatment visual acuities; furthermore, it is not clear whether enucleation was counted as an adverse visual event.

Both Packer et al¹⁴ and Char et al²⁰ reported baseline and treatment factors associated with posttreatment visual acuity in their study populations. Packer et al found that the presence of diabetes increased the risk of an adverse visual outcome; they speculated that the diabetic vasculature had increased susceptibility to radiation damage. A recent report by Gragoudas et al³⁵ found that patients with diabetes were at greater risk of radiation maculopathy and papillopathy after proton beam irradiation. Char et al reported that tumors more than 3 mm from the optic nerve and fovea and tumors less than 6 mm in thickness had better visual acuity after treatment than larger tumors and tumors within 3 mm of the nerve or fovea. It should be noted that distance between the tumor and a critical eye structure is highly correlated with radiation dose to the structure; distance from the tumor to the FAZ in the COMS models may be replaced with the radiation dose to the FAZ with little effect on the fit of the model or on the estimates of relative risk associated with other baseline characteristics. Overall, COMS results regarding risk factors are consistent with the findings of previous studies.

The impact that treatment and subsequent loss of vision in one eye have on visual functioning and quality of life is currently under investigation in a parallel study to the COMS.³⁶ In 2003, the COMS will have completed at least 5 years of follow-up for all 1317 patients enrolled in the trial for medium-sized tumors, and initial results regarding survival and quality of life after treatment with enucleation versus I¹²⁵ brachytherapy are anticipated by that time.

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Kellogg Eye Center, Ann Arbor, MI. **Principal Investigator:** Andrew K. Vine, MD; **Clinic Coordinator:** Julie M. Willis, LPN, COMA; **Ophthalmologists:** Ron M. Kurtz, MD; Scott R. Sneed, MD (1989-1992); **Radiation Oncolo-**

gists: James A. Hayman, MD; Allen S. Lichter, MD; Cornelius J. McGinn, MD; John M. Robertson, MD (1992-1995); Sonja Schoepel, MD (1988-1991); Richard Diaz, MD (1986-1987); **Radiation Physicists:** Randall Ten Haken, PhD; Antoinette V. Thompson, MS (1995-1997); **Echographer:** Kathleen A. Meyer, RDMS; **Photographers:** Csaba Martonyi; Sally Ann Stanley; Fran McIver (1989-1993); **Ophthalmic Pathologists:** Victor Elner, MD, PhD; Reimer J. Wolter, MD (1988-1993); Marilyn Kincaid, MD (1986-1987).

Emory Eye Center, Atlanta, GA. **Principal Investigator:** Paul Sternberg, Jr, MD; Travis A. Meredith, MD (1986-1990); **Clinic Coordinator:** Jayne M. Brown; **Ophthalmologists:** Thomas M. Aaberg, Jr, MD; Thomas M. Aaberg, Sr, MD; Antonio Capone, Jr, MD; Ted H. Wojno, MD; **Radiation Oncologists:** James W. Keller, MD; Jerome Landry, MD; Ian R. Crocker, MD (1992-1995); John McLaren, MD (1986-1993); **Radiation Physicists:** Elizabeth Butker, MS; Kathleen A. Klee, PhD; Christopher A. Aguilera, MS (1991-1997); David Strongosky, MMSc (1987-1988); **Visual Acuity Examiners:** Deborah K. Gibbs, COT; Ann Fremstad, COT (1991-1994); Linda T. Curtis, COA (1986-1993); **Echographers:** Rhonda G. Waldron, COMT; Alice Ward (1986-1990); **Photographers:** James Gilman, CRA; Robert A. Myles; Ray Swords, CRA; Chris Snipes (1987-1987); **Ophthalmic Pathologists:** Hans Grossniklaus, MD; F. Phinizy Calhoun, Jr, MD (1988-1993); Wallace Campbell, MD (1987-1989).

Piedmont Hospital, Atlanta, GA. **Principal Investigator:** James H. Frank, MD; William H. Jarrett II, MD (1986-1998); **Ophthalmologists:** Robert L. Halpern, MD; William S. Hagler, MD (1987-1998); **Radiation Oncologists:** Norman Jones, MD; Fred P. Schwaibold, DO; **Radiation Physicists:** Karol J. Wynn, CMD; Peter Mondalek (1987-1989); **Visual Acuity Examiners:** Christine E. Cooper; Gail A. Degenhardt, COA; Angela R. Muncey, COA; Erin K. Hulsey, COA (1996-1998); Debbie H. Lee, COA (1990-1998); Lisa Kailey, COA (1996-1997); Susan N. McCart (1991-1995); Gail A. McCoy (1991-1995); Jon Aaron, COT (1986-1994); David S. Bolerjack, COA (1990-1991); **Photographers:** Kenneth D. Thompson; Kelly K. Wilson (1992-1996); Bambi P. Robinson, CRA (1991-1993); Ben Baxter (1987-1989).

The Wilmer Ophthalmological Institute, Baltimore, MD. **Principal Investigator:** Andrew P. Schachar, MD; **Clinic Coordinators:** Warren T. Doll, Jr, COA; Ellen F. Greenberg, COT; Mike Hartnett, COT; Marguerite Alexander, RN, BSN (1986-1995); Catherine S. Sackett, RN, CANP (1989-1993); Susan Kaiser (1988-1989); Jane Ebert Torano (1987-1987); **Ophthalmologists:** Morton F. Goldberg, MD; Lenard H. Hammer, MD; Nicholas T. Iliff, MD; **Radiation Oncologists:** Ding J. Lee, MD, PhD; Mi-Son Chun, MD (1991-1993); **Radiation Physicists:** Juan F. Jackson; Wing-Chee Lam, PhD (1987-1992); Joseph Ianitto (1987-1990); **Visual Acuity Examiner:** Linda T. Pollack, CO (1988-1993); **Echographer:** Cathy DiBernardo, RN, RDMS; **Photographers:** Dennis Cain; David Emmert; Terry George.

Retina Associates of Cleveland, Beachwood, OH. **Principal Investigators:** Z. Nicholas Zakov, MD; Thomas A. Rice, MD (1987–1997); **Clinic Coordinators:** Geraldine Daley, CRA; Dina Preseren (1988–1991); Kristen Dempsey (1987–1989); **Ophthalmologists:** Michael A. Novak, MD; Scott D. Pendergast, MD; Lawrence J. Singerman, MD; Daniel Weidenthal, MD; Hernando Zegarra, MD; John H. Niffenegger, MD (1995–1998); James H. Antoszyk, MD (1994–1996); **Radiation Oncologist:** Donald Shina, MD; **Radiation Physicists:** Ned Began, MS; Kunjan Pillai, MS; **Visual Acuity Examiners:** Mary A. Ilc, COT; Connie S. Keller, COA; Donna Knight, COT; Stephanie A. Schura, COA; Kimberly A. Tilocco, COA; Kathy Coreno (1989–1994); **Echographers:** Pamela Rowe, CRA; Kelly Smith, CRA, COT (1989–1994); **Photographer:** Sheila Smith-Brewer, CRA, COMT.

Schepens Retina Associates, Boston, MA. **Principal Investigator:** Clement Trempe, MD; **Clinic Coordinator:** Zena F. Nies; **Ophthalmologists:** John Weiter, MD; Jay S. Duker, MD (1990–1993); **Radiation Oncologist:** David E. Wazer, MD (1989–1994); **Radiation Physicist:** Kenneth Ulin, PhD (1988–1994); **Visual Acuity Examiners:** Gerald Friedman, OD; Rodney Immerman, OD; **Echographers:** Alex Jalkh, MD; Fernando Acosta, MD (1990–1993); Miguel A. Quiroz, MD (1989–1990); Fadi Nasrallah, MD (1987–1989); **Photographers:** Dennis C. Donovan; Thomas O'Day (1987–1995); Andrew Rosenblum (1986–1989); Cheryl Weber (1986–1988).

University of Virginia, Charlottesville, VA. **Principal Investigator:** Brian P. Conway, MD; **Clinic Coordinators:** Mary Jonni Henofer, RN, CRNO; Diana D. Robertson, LPN COA (1989–1989); **Ophthalmologists:** James S. Tiedeman, MD, PhD; Eleanore M. Ebert, MD, MPH (1993–1995); **Radiation Oncologists:** William C. Constable, MD; James M. Larner, MD; Michela Caruso, MD (1992–1994); Cynthia Spaulding, MD (1989–1992); **Radiation Physicists:** Gilbert D. Glennie, ME, Biomed; Janelle A. Molloy, PhD; Alan Aqualino, PhD (1989–1996); B. Ann Normansell, ME, ABR (1990–1992); **Visual Acuity Examiners:** James S. Chisholm, COT; Patsy Overstreet, RN, COT; Melanie Smith, COT (1994–1995); Karen L. Summerville, COMT (1994–1995); L. Sharon Hoyle, COMT (1992–1994); Ellen T. Murphy, AS, COA (1993–1994); **Photographers:** Jane G. Fleming; James Scott.

Northwestern University, Chicago, IL. **Principal Investigator:** Lee M. Jampol, MD; James Puklin, MD (1987–1989); **Clinic Coordinators:** Beth Chiappetta, RN; Jill M. Koecher; Mimi S. Mansfield (1996–1997); Gail Daubert, RN (1987–1993); **Ophthalmologists:** Marianne Cowley, MD; Mark Daily, MD; Richard G. Gieser, MD; Gary S. Lissner, MD; Robert Schroeder, MD; David V. Weinberg, MD; **Radiation Oncologists:** Bharat B. Mittal, MBBS; William Small, Jr, MD; Ramananda Shetty, MD (1986–1996); **Radiation Physicists:** Patricia Johnson, DSc; Alan G. Kepka, PhD ABR; **Visual Acuity Examiners:** Edgar S. Perez, OD; Zuzanna Strugala, COA, MA; Jeff M. Scurry (1993–1995); Barbara A. Tallman, COT (1991–1992); Michael Giese, OD (1987–1991); **Echographers:** Robert A. Levine, MD; William Myers, MD (1986–1993); **Photographers:** John M. Gerty, CRA; Alexander Habib; Leonard S.

Richine (1988–1998); Gloria J. Valadez (1995–1996); Karen Parque (1988–1993); Ned Bezinovich (1988–1992); **Ophthalmic Pathologist:** Richard O'Grady, MD; **Following Oncologists:** Steven B. Newman, MD; John M. Shaw, MD.

University of Illinois, Chicago, IL. **Principal Investigator:** Norman P. Blair, MD; **Clinic Coordinators:** Christine Mullahy, COT; Dorris Brown-Hutchins (1988–1998); Moira Keating (1986–1987); **Ophthalmologist:** Kenneth I. Resnick, MD (1990–1999); **Radiation Oncologist:** Edwin Liebner, MD (1986–1995); **Radiation Physicist:** Richard Haas (1987–1995); **Visual Acuity Examiners:** Andrew Cross, COT; Dorothe Ernest, MA, ACSW (1989–1990); **Echographers:** Carolyn Anderson, MD (1993–1995); Leonard Kut, MD (1987–1992); **Photographer:** Norbert Jednock.

Cleveland Clinic Foundation, Cleveland, OH. **Principal Investigator:** Froncie A. Gutman, MD; **Clinic Coordinators:** Laura Holody, COA; Susan E. Lichterman, RN; Tina E. Kiss, COT, CCRA (1994–1997); Vivian Tanner, COT (1987–1993); **Ophthalmologists:** Jill A. Foster, MD; Umberto Albanese, MD (1991–1995); John V. Linberg, MD (1990–1991); John S. Ambler, MD (1989–1990); **Radiation Oncologists:** Robert M. Fine, MD (1993–1998); Jerrold P. Saxton, MD (1990–1997); Rashad El Dabhi, MD (1988–1990); **Radiation Physicists:** Eric J. Lee, MSc (1995–1997); Patrick Higgins, PhD (1987–1994); Debbie A. Jenkins, PhD (1993–1994); Claudio H. Sibata, PhD (1989–1990); **Visual Acuity Examiner:** Susannah Hanson, CO (1987–1997); **Echographers:** Karen A. King, COT; LuAnne Sculley, COMT, RDMS; Cate A. Reinhard (1993–1996); Louise Berlin (1987–1989); **Photographers:** Deborah Ross, CRA; Michael P. Kelly (1995–1997); Pamela J. Vargo (1995–1996); Roberta Abram, CRA (1986–1990); **Ophthalmic Pathologist:** Careen Lowder, PhD, MD.

Ohio State University, Columbus, OH. **Principal Investigator:** Frederick H. Davidorf, MD; **Clinic Coordinators:** Cynthia S. Taylor; Susan E. Cornetet (1991–1993); Jean Bildsten, RN (1986–1990); **Ophthalmologists:** Robert B. Chambers, DO; Joseph J. Fammartino, MD; **Radiation Oncologists:** Constance J. Bauer, MD; Sheila E. Hodgson, MD; Subir Nag, MD; **Radiation Physicists:** Christos Kanelitsas, PhD; Nina Samsami, PhD; Charles F. Wild, PhD (1993–1995); Cheng M. Su, PhD/ABR (1989–1991); **Visual Acuity Examiners:** Chhanda Chaudhuri, COA; Jill D. Milliron, COA; Jerilyn G. Perry, COT, ABO; Nanci J. Cover, COA (1992–1995); Gail Shortlidge (1987–1995); **Echographer:** Paula Andrzejewska (1993–1993); **Photographers:** Michael J. Keating; Scott J. Savage, EMT-A; Debra Weisenburger, CRA (1989–1992); R. Michael Clark, CRA (1986–1990); **Following Oncologist:** Gregory J. LaValle, MD.

Texas Retina Associates, Dallas, TX. **Principal Investigator:** Dwain Fuller, MD; **Clinic Coordinators:** Sally A. Arceneaux, COST, COA; Jean Arnwine; Theresa Anderson, COA (1986–1997); **Ophthalmologists:** Gary Edd Fish, MD; William Snyder, MD; **Radiation Oncologist:** Donald Schwarz, MD; **Radiation Physicists:** Laura A. Berey, MS; William F. Gagnon, PhD/ABR; David B. Hammond, ABMP; Gregory McDaniel, PhD; Gene H. Sowell, MS

(1995–1997); **Visual Acuity Examiners:** Cynthia Nork (1987–1995); Denise Dalrymple (1987–1991); Diane Conway (1987–1988); **Echographers:** Beth A. Pieczynski, COA (1991–1998); Randal Straach (1987–1995); **Photographers:** Hank Aguado; Bob H. Boleman; Richard E. Jones; John G. King; Richard Evans, CRA (1988–1989); Kevin Rose (1988–1989); Ruth Picchiottino (1986–1987); **Ophthalmic Pathologist:** Martha Luckenbach, MD (1987–1989).

Porter Adventist Hospital/Centura Health, Denver, CO. Principal Investigator: Kenneth R. Hovland, MD; **Clinic Coordinator:** Suzette Compton, RN; **Ophthalmologists:** Gary J. L. Foster, MD; Douglas L. Holmes, MD; David W. Johnson, MD; Stephen T. Petty, MD; John Pope, Jr, MD; Matthew W. Wilson, MD; John D. Zilis, MD; Richard O. Hammond, MD (1991–1995; deceased); **Radiation Oncologists:** Seth D. Reiner, MD; David S. Shimm, MD; Wendy M. Nekritz, MD (1997–1998); David P. Schreiber, MD (1989–1997); Amanda J. Story, MD (1991–1992); **Radiation Physicists:** Kari L. Cann, MS; Gary W. Douglas, MSc; Michael J. Bailey, MS (1990–1996); J. Donald Russell, MS/ABR (1991–1992); Gregory Burns, MS (1989–1990); **Echographers:** David Beers; Sandra R. Sall; **Photographers:** Lawrence Disney, COA; Martin Lopez, COT (1989–1990); **Ophthalmic Pathologist:** Robert Keyser, MD (1989–1993).

Duke Eye Center, Durham, NC. Principal Investigator: Jonathan J. Dutton, MD, PhD; **Clinic Coordinators:** Kim K. Peddle, COT; Cynthia Lyons, AA (1994–1997); Barbara A. Schuh, COT (1990–1994); Ellen S. Feuer, COT (1989–1990); Marianne Heston, CMA, COT (1986–1989); **Ophthalmologist:** Christopher M. DeBacker, MD; **Radiation Oncologists:** Mitchell S. Anscher, MD; Carol A. Hahn, MD; Edward Halperin, MD; Lawrence B. Marks, MD; Gregory S. Sibley, MD (1996–1998); Jeffery D. Morton, MD (1995–1995); Gustavo Montana, MD (1987–1994); Kenneth Leopold, MD (1988–1993); Angel E. Torano, MD (1992–1993); **Radiation Physicists:** Gunilla C. Bentel, RN, RTT; Thaddeus Samulski, PhD; Kenneth J. Weeks, PhD; Gerard Honore (1987–1988); **Photographers:** Ruth Schirmer, CRA (1987–1997); Beth Ann Benetz, CRA (1989–1994); Nancy E. Knight, CRA, COA (1991–1994); **Ophthalmic Pathologists:** Gordon Klintworth, MD, PhD (1986–1992); Alan Proia, MD, PhD (1987–1992).

Hermann Eye Center, Houston, TX. Principal Investigator: Richard S. Ruiz, MD; **Clinic Coordinators:** Dalia Vargas; Robert W. Johnson, COA, RDO (1992–1993); Maria DeLaGarza (1988–1992); Georgia Triplett, OA (1992–1992); Walter (Gus) Kohn (1987–1989); **Ophthalmologists:** Graham P. Avery, MD; William L. Decker, MD; Judianne Kellaway, MD; H. Michael Lambert, MD; Paul C. Salmonsens, MD; Jay B. Stallman, MD (1993–1996); Debra J. Shetlar, MD (1993–1994); **Radiation Oncologists:** Nora A. Janjan, MD; William Morrison, MD (1987–1996); Joseph Kong, MD (1987–1987); **Radiation Physicist:** Otto Zeck, PhD; **Echographers:** Thomas C. Prager, PhD; Yali Zou, MD; **Photographers:** Charles B. Smith; Teddy J. Horton (1987–1989).

Midwest Eye Institute, Indianapolis, IN. Principal Investigator: John T. Minturn, MD; **Clinic Coordinators:**

Donna J. Agugliaro, RN, BSN; Margaret B. Kirby, RN (1989–1992); Kimberly A. Sharkey, CST (1989–1989); **Ophthalmologists:** Ronald T. Martin, MD; William R. Nunery, MD; **Radiation Oncologist:** Peter Garrett, MD; **Radiation Physicists:** Ronald E. Berg, PhD; John Kent, MS; **Photographers:** Carolyn S. Lamb; Thomas C. Meador (1997–1997); Kenneth D. Phelps (1995–1996); William T. Bussell (1995–1995); Linda Huber, CRA (1989–1995); **Ophthalmic Pathologist:** Jose M. Bonnin, MD.

The University of Iowa, Iowa City, IA. Principal Investigator: Thomas A. Weingeist, MD, PhD; **Clinic Coordinators:** Connie Fountain, COT; Marcia Griffin; **Ophthalmologists:** H. Culver Boldt, MD; Keith D. Carter, MD; Jeffrey A. Nerad, MD; Edwin M. Stone, MD, PhD; Christopher F. Blodi, MD (1987–1992); **Radiation Oncologists:** David H. Hussey, MD; B-Chen Wen, MD; Antonio Vigliotti, MD (1987–1991); John Staples, MD (1988–1989); **Radiation Physicists:** Edward Pennington, ScM; Womah S. Neeranjun, MS (1994–1997); Shirish K. Jani, PhD, DABR (1987–1993); **Echographers:** Tom Fisher; Laura Warner; Karl C. Ossoinig, MD (1986–1997); Vicki Crosser (1986–1989); Kamel Itani (1986–1987); **Photographers:** Edward Heffron, MA; Stefani V. Karakas, CRA; Randall E. Verdick; **Ophthalmic Pathologist:** Robert Folberg, MD; **Following Oncologists:** Gerald H. Clamon, MD; Susan A. Kambhu, MD (1996–1998).

Estelle Doheny Eye Foundation, Los Angeles, CA. Principal Investigators: A. Linn Murphree, MD; Peter E. Liggett, MD (1986–1991); Stephen J. Ryan, MD (1986); **Clinic Coordinators:** Margaret Padilla; A. Frances Walonker, CO, COMT; Beth Quillen-Thomas, COMT (1987–1991); **Ophthalmologists:** Kenneth R. Diddie, MD; Jennifer I. Lim, MD; Leroy C. McNutt, MD; Brent C. Norman, MD; Donald A. Frambach, MD (1991–1998); Sunil S. Patel, MD, PhD (1997–1998); Michael A. Singer, MD (1993–1997); **Radiation Oncologists:** Deirdre Cohen, MD; Zbigniew Petrovich, MD; Daphne Palmer, MD (1988–1990); **Radiation Physicists:** Melvin A. Astrahan, PhD; Gary Luxton, PhD; **Visual Acuity Examiner:** Maria Trujillo (1987–1987); **Echographer:** Ronald L. Green, MD; **Photographers:** Tracy Nichols, CRA; Ben R. Serar (1990–1993); Mark G. Williams, CRA (1990–1993); Stephen Delgado (1987–1990); Michon Rozier (1987–1989); Jan Johnson (1987–1987).

Jules Stein Eye Institute, Los Angeles, CA, and Southern California Permanente Group, Panorama City, CA. Principal Investigator: Bradley R. Straatsma, MD; **Clinic Coordinators:** Robert Caldwell, PA; Karen Baranick, COA (1993–1995); Barbie S. Pinaire, COA (1990–1990); Margaret G. Haslop, COA (1987–1989); **Ophthalmologists:** Robert E. Engstrom, Jr, MD; Man M. Singh Hayreh, MD; Hector Sulit, MD; **Radiation Oncologists:** C. Michelle Burnison, MD; Robert G. Parker, MD; Michael Selch, MD (1987–1989); **Radiation Physicists:** James Smathers, PhD; Robert E. Wallace, MEng, PhD, DABR; F. Eugene Holly, PhD (1990–1993); **Visual Acuity Examiners:** Lisa A. Barnhart, OD; Melissa Chun, OD; Burton Krell, OD (1988–1995; deceased); **Echographers:** Marci L. Fishman, RDMS; Barry Kerman, MD; Joel N. Moral, COT; Nancy London, COA (1986–1996); Ron Smith, COT (1991–

1993); Lynn Wynbrandt (1987–1989); **Photographers:** Michael J. Heneghan; Dennis Thayer; Orly Catz (1996–1997); Anne P. Bolton (1991–1995); Don Allen (1989–1990); Joan Greenspan (1988–1989); **Ophthalmic Pathologists:** Ben J. Glasgow, MD; Robert Y. Foos, MD (1987–1994).

University of Wisconsin, Madison, WI. Principal Investigator: Suresh R. Chandra, MD; **Clinic Coordinators:** Margo S. Blatz; Jennie R. Perry; Betty Lewis (1987–1998); Jennifer L. Miller (1990–1991); Lyn Stewart (1987–1990); **Ophthalmologists:** Justin L. Gottlieb, MD; Bradley Lemke, MD; Mark J. Lucarelli, MD; Timothy W. Olsen, MD; Richard Dortzbach, MD (1986–1998); Frank Myers, MD (1987–1998); Thomas Stevens, MD (1986–1995); **Radiation Oncologists:** Minesh P. Mehta, MBChB/ACR; Richard Steeves, MD, PhD; **Radiation Physicists:** Bhudatt R. Paliwal, PhD; Bruce Thomadsen, PhD, Siamak Shahabi, PhD (1987–1991); **Visual Acuity Examiners:** Jim N. Henry, COT; Guy Somers, RN; Helen Lyngaas (1993–1998); Diane Quackenboss, RN (1987–1995); **Echographer:** Laura Gentry, COA; **Photographers:** Robert Harrison; Gene Knutson; Hugh D. Wabers; Michael Neider (1987–1995).

North Shore University, Manhasset, NY, and New York Eye and Ear Infirmary, New York, NY. Principal Investigator: Paul T. Finger, MD; **Clinic Coordinators:** Paula A. Muir; Cynthia Pearlberg (1987–1996); **Ophthalmologist:** Samuel Packer, MD; **Radiation Oncologists:** Anthony M. Berson, MD; Jay Bosworth, MD; Manjeet Chadha, MD (1993–1998); Julianna Pisch, MD (1993–1998); David L. Sherr, MD (1996–1998); Louis Potters, MD (1992–1994); Daniel DeBlasio, MD (1988–1991); **Radiation Physicists:** Nokul Panigrahi, PhD; Richard Riley, PhD, ABR; Andrzej Szechter, PhD; Shantilata Mishra, PhD, DABR (1992–1996); Tony Ho, MS (1987–1991); Willie Vahid Jalayer, PhD (1991–1991); **Visual Acuity Examiner:** Lori R. Reminick, OD (1997–1997); **Photographers:** Nancy Gonzalez, COA; James W. Perry; Jon Wilder; Randall Honeywell (1993–1994); Roy E. Tuller, RBP (1989–1990); Michael McCray (1987–1989); **Ophthalmic Pathologist:** Steven A. McCormick, MD.

Bascom Palmer Eye Institute, Miami, FL. Principal Investigator: Timothy G. Murray, MD; Donald H. Nicholson, MD (1987–1995); **Clinic Coordinators:** Nicole Ciciarelli, CST; Joanka M. LoBracco; Madeline L. Del Calvo (1992–1995); Marianela Castellanos (1987–1994); Ana M. Soler (1991–1993); **Ophthalmologists:** Leonard Berg, MD (1990–1995); Harold J. Berger, MD (1990–1995); William R. Burks, MD (1989–1995); Howard A. Doyle, Jr, MD (1990–1995); Mary Lou Lewis, MD (1990–1995); Alfred H. Rivera, MD (1990–1995); Patrick E. Rubsamen, MD (1991–1995); David R. Simon, MD (1990–1995); David A. Snyder, MD (1990–1995); Barry S. Taney, MD (1990–1995); **Radiation Oncologists:** Arnold M. Markoe, MD, ScD; James Schwade, MD (1988–1994); Beatriz E. Amendola, MD (1990–1991); Bruce Horowitz, MD (1987–1988); Abdon Medina, MD (1988–1988); **Radiation Physicists:** Xiaodong Wu, PhD; Jeffrey A. Fiedler, MSc (1992–1996); Pavel Houdek, PhD (1987–1995); Vincent J. Pisciotta, MS (1988–1995); **Visual Acuity Examiners:** Nury Delgado; Nidia Y. Rosado-Guzauskas; Rita D. Swoopes; Mailyn Gonzalez (1993–1997); Esther Pastrana, COT (1994–

1996); June Thompson (1993–1994); Deyanira I. Serrano (1992–1993); Steven C. Deen, OD (1988–1992); Donald R. Faimon, OD (1990–1992); David J. Kowal, OD (1988–1990); Mark D. McClintock, OD (1988–1990); Mark Liebetreu, OD (1988–1989); Edward Attaway (1987–1988); Steven Holbrook (1987–1988); **Echographers:** Fiona Ehli; J. Randall Hughes; E. Kym Gendron (1987–1999); Sandra Frazier Byrne (1987–1994); **Photographers:** Ditte Hess; Isabel Rams, CRA; Ross Jarrett (1988–1992); **Ophthalmic Pathologists:** Victor T. Curtin, MD; Robert H. Rosa, Jr, MD; **Following Oncologist:** Lynn G. Feun, MD.

Medical College of Wisconsin, Milwaukee, WI. Principal Investigators: William F. Mieler, MD; Gary W. Abrams, MD (1985–1986); **Clinic Coordinators:** Sharon A. Rekow; Prudence Helt, LPN (1986–1988); **Ophthalmologist:** Gerald J. Harris, MD; **Radiation Oncologists:** Beth Erickson, MD; Colleen A. Lawton, MD; Maurice Greenberg, MD (1986–1991); **Radiation Physicists:** Michael T. Gillin, PhD; Katherine S. Albano, MS/ABR (1994–1998); Darwin L. Zellmer, PhD (1991–1995); **Visual Acuity Examiner:** Dale T. Jurkiewicz (1989–1993); **Echographer:** Michael Lewandowski, RDMS (1986–1996); **Photographers:** James R. Phillips, AS; Walter Wipplinger; Mary Nauertz-Ritchie (1987–1991); Jim L. Walters (1989–1989); **Ophthalmic Pathologist:** James Caya, MD (1986–1988).

Montreal Hospitals, Montreal, Canada. Principal Investigator: Christine Corriveau, MD; **Clinic Coordinators:** Liette Bousquet, COA; Suzanne Casavant, RN; Margaret Gagnon, RN (1995–1998); Darlene O'Connor (1989–1993); **Ophthalmologists:** Mikael A. Sebag, MD; Marie Carole Boucher, MD (1990–1998); Jean Deschenes, MD (1989–1998); Francois Codere, MD (1989–1997); **Radiation Oncologists:** Michel Gelinas, MD; Pierre Rousseau, MD; Mariam Yassa, MB, BCH (1989–1998); Claudine Theriault, MD (1989–1993; deceased); **Radiation Physicists:** Noel Blais, PhD; Michael D. C. Evans, MSc, FCCPM (1990–1998); Brenda G. Clark, PhD, MCCPM (1993–1997); Real Besner, MS (1990–1994); **Visual Acuity Examiners:** Claire Choquette, OC; Deborah M. Williams, COT (1990–1998); **Echographers:** Magdi M. Mansour, MD, MSc (1993–1998); Diane Chialant, RN (1989–1992); **Photographers:** Marc J. Blouin, OA; Lilianne L. M. Marcon, COA (1993–1998); Rosario Bate, BN, CRA, COT (1990–1992); **Ophthalmic Pathologists:** Guy S. Allaire, MD; Miguel N. Burnier, MD, PhD (1994–1997); Seymour Brownstein, MD, FRCSC (1990–1992).

Touro Infirmary, New Orleans, LA. Principal Investigator: Gerald Cohen, MD; **Clinic Coordinators:** Sandra W. Palao, CST; Susy A. Steib (1989–1992); **Ophthalmologist:** Kurt A. Gitter, MD; **Radiation Oncologist:** Mary Ella Sanders, MD; **Radiation Physicists:** Lawrence J. Metevier, MS; Donald G. Jackson (1990–1992); Howard V. Kavanaugh, MS (1990–1992); **Echographer:** Mary E. Smith, RDMS (1990–1995).

Tulane University, New Orleans, LA (1989–1995). Principal Investigator: Barrett Haik, MD (1989–1995); **Clinic Coordinators:** Elaine M. Eckols, COA (1990–1995); Carolyn A. Bushaw, RN (1989–1991); **Ophthalmologist:** Zeynel A. Karcioğlu, MD (1989–1993); **Radia-**

tion Oncologist: David J. Bush, MD (1989–1994); **Radiation Physicist:** George Meckstroth, PhD (1990–1995); **Visual Acuity Examiner:** Alan Faulkner, MD; **Photographers:** Frank J. Saucier (1990–1995); Rebecca Thiele, CRA (1990–1995).

Cornell University, New York, NY. Principal Investigator: David H. Abramson, MD; **Clinic Coordinators:** Diane M. Murphy; Indira S. Rollins, CST; Katherine S. Budinger, COA (1995–1997); Camille A. Servodidio, RN, MPH, CRNO (1986–1997); Stewart P. Martin, CRA (1993–1994); Agnese Romanella-Weil, RN, BSN, CRNO (1989–1992); **Ophthalmologist:** Mary E. Mendelsohn, MD (1994–1996); **Radiation Oncologists:** Hassan I. Aziz, MD; Beryl McCormick, MD; Karen D. Schupak, MD (1993–1997); Daniel Fass, MD (1987–1993); **Radiation Physicists:** Raj P. Bhalla, PhD, ABR; Patrick J. Harrington; Soutung Chiu-Tsao, PhD (1987–1990); **Photographers:** Dorothy A. Fong, COT; Rob Wolfson, CRA; **Ophthalmic Pathologists:** Winston G. Harrison, MD; Sarah S. Frankel, MD (1990–1991).

McGee Eye Institute, Oklahoma City, OK. Principal Investigator: Reagan H. Bradford, Jr, MD; **Clinic Coordinator:** Jamie Johnson, LPN; Janie M. Shofner, COA, CCRA (1990–1998); **Ophthalmologists:** P. Lloyd Hildebrand, MD; Sumit K. Nanda, MD; Scott C. Sigler, MD; **Radiation Oncologists:** Carl R. Bogardus, Jr, MD (1989–1995); Robert C. Gaston, DO; Elizabeth J. Syzek, MD (1995–1998); **Radiation Physicists:** B. Wally Ahluwalia, PhD; David E. Raeside, PhD (1989–1995); **Echographer:** Jaime Rodriguez; **Photographer:** Russell D. Burris, CRA, COT.

Wills Eye Hospital, Philadelphia, PA (1992–1995). Principal Investigator: Jerry A. Shields, MD; **Clinic Coordinators:** Diane McDonald, COT; Margaret C. Morley; **Ophthalmologists:** Patrick V. De Potter, MD; Carol L. Shields, MD; **Radiation Oncologist:** Luther W. Brady, MD; **Radiation Physicists:** Lester Tripp, MSc; Patrick Glennon, MSc (1992); **Echographer:** Queen Warrick, COA; **Photographer:** Matthew Weiner, CRA; **Ophthalmic Pathologist:** Ralph C. Eagle, Jr, MD.

Retina-Vitreous Consultants, Pittsburgh, PA. Principal Investigators: Karl R. Olsen, MD; Jeffrey S. Rinkoff, MD (1988–1996); **Clinic Coordinators:** Barbara J. Bahr, LPN; Donna L. Green, RN, CRNO; **Ophthalmologists:** Randall L. Beatty, MD; Robert L. Bergren, MD; Gary Weinstein, MD; Thierry Verstraeten, MD (1992–1995); Scott R. Hobson, MD (1991–1994); John S. Kennerdell, MD (1988–1991); **Radiation Oncologists:** Melvin Deutsch, MD; John C. Flickinger, MD; **Radiation Physicists:** Kurt G. Blodgett, MS; Andrew Wu, PhD (1988–1994); Virgil E. Yoder, MS (1988–1989); **Visual Acuity Examiners:** Lynn A. Wellman; Linda A. Wilcox, COA; Kimberly S. Flook, COA (1992–1993); Beverly J. Keenen (1991–1992); Catherine Jaborsky (1988–1989); **Photographers:** Alan F. Campbell, CRA; David Steinberg, CRA; Gary Vagstad, CRA; Mary M. Good (1993–1993); Kim A. Meltinger, COA (1991–1991).

Casey Eye Institute and Devers Eye Clinic, Portland, OR. Principal Investigator: David J. Wilson, MD; Robert C. Watzke, MD, FACS; Richard G. Chenoweth, MD

(1987–1996); **Clinic Coordinators:** Shirley Ira, COT; Anna Jane Leigh, COT (1989–1992); Elaine Lorimer (1987–1989); **Ophthalmologists:** Roger A. Dailey, MD; Richard F. Dreyer, MD; Diana L. Habrich, MD; Michael Klein, MD; Colin Ma, MD; Joseph Robertson, MD; John L. Wobig, MD, FACS; Frederick Fraunfelder, MD (1987–1995); **Radiation Oncologists:** Harper D. Pearce, MD; Kenneth R. Stevens, MD; Mark F. Schray, MD (1992–1995); Roland Ostlund, MD (1987–1992); Juan V. Fayos, MD (1989–1989); **Radiation Physicists:** Raymond Fry, MS; Robert J. Miller, MS/ACR (1990–1995); Douglas Jones, PhD (1987–1993); **Visual Acuity Examiners:** Ellen F. Redenbo, COA; Julie Arends, COMT (1987–1995); Lori Rusch (1987–1989); **Echographers:** Mark Evans; Patrick B. Rice, CRA (1991–1997); Thomas Talbot, MD (1986–1991); **Photographers:** Howard G. Daniel (1990–1995); Fernando Corrada (1988–1989).

Mayo Foundation, Rochester, MN. Principal Investigator: Dennis M. Robertson, MD; **Clinic Coordinators:** Margaret J. Rusczyk, CCRA; Mary Schumann, COMT; **Ophthalmologists:** Helmut Buettner, MD; John Pach, MD (1988–1993); **Radiation Oncologists:** Ivy A. Petersen, MD; Paula Schomberg, MD; Scott L. Stafford, MD; John D. Earle, MD (1986–1996); **Radiation Physicists:** Robert W. Kline, PhD; Peter Yeakel, RTT; **Visual Acuity Examiners:** Barbara Eickhoff, COMT; Deborah Miller, COMT; **Photographers:** Thomas Link; Jay Rostvold; **Ophthalmic Pathologist:** R. Jean Campbell, MD (1988–1998); **Following Oncologists:** Kurt W. Carlson, MD; Taylor J. Hays, MD.

Associated Retinal Consultants, Royal Oak, MI. Principal Investigator: Raymond R. Margherio, MD; **Clinic Coordinators:** Kristi Cumming, RN, MSN; Patricia Mantrey, RN; Beth Mitchell, RN (1989–1998); Sheri McCurley, RN, BSN (1986–1989); **Ophthalmologists:** Bruce R. Garretson, MD; Alan R. Margherio, MD; Michael T. Trese, MD; Philip J. Ferrone, MD (1996–1998); Patrick L. Murphy, MD (1987–1995); **Radiation Oncologists:** Gary S. Gustafson, MD; Alvaro Martinez, MD; Richard Matter, MD; **Radiation Physicists:** Gregory Edmundson, RT (T); Elizabeth C. Mele, MS; Nancy R. Rizzo, MS (1992–1992); **Visual Acuity Examiner:** Virginia S. Regan, RN, BSN; **Echographers:** Laurie K. Lau-Sickon, RDMS; Andrea P. Nielsen, OD; William Park, OD (1987–1998); **Photographers:** Craig Bridges, CRA; Rachel E. Falk; Frances McIver; Patricia Streasick; Lynette D. Szydlowski, COT; John L. Johnson (1992–1994); Deborah Sorenson (1987–1992); Jeffrey Sobel, CRA (1987–1990); **Following Oncologists:** David A. Decker, MD; Ishmael Jaiyesimi, MD.

Washington University, St Louis, MO. Principal Investigators: Lucian Del Priore, MD (1995–1998); Morton E. Smith, MD (1986–1995); **Clinic Coordinators:** Mary V. Gaines; Lori Ann Clark, COT (1994–1996); Floyce Scherrer, RN (1987–1987); **Ophthalmologists:** Philip L. Custer, MD; Henry J. Kaplan, MD; Lawrence Schoch, MD (1987–1989); **Radiation Oncologists:** Hsiu-San Lin, MD, PhD (1989–1995); Venkata Devineni, MD (1987–1994); **Radiation Physicists:** Jeffrey Williamson, PhD (1991–1995); Eric Slessinger, MS (1987–1994); **Echographer:** William Hart, MD; **Photographers:** Rhonda Curtis, COT; Gregory C. Hoffmeyer (1996–1997).

University of Texas, San Antonio, TX. **Principal Investigator:** Wichard van Heuven, MD; Gary Cowan, MD (1986–1988); **Clinic Coordinators:** Leticia S. Leija, RDMS; Carolyn G. Torres, LVN (1993–1996); Jacalyn McAdam, ARDMS (1990–1993); Debbie A. Schifanella (1991–1993); Judy Attaway (1987–1991); Barry M. Byrne (1987–1989); **Ophthalmologists:** Brian B. Berger, MD; Steven Chalfin, MD; David E. Holck, MD; Lt. Colonel Donald A. Hollsten, MD; Bailey L. Lee, MD; Lina M. Marouf, MD; Kenneth L. Piest, MD; James W. Speights, MD; Christopher T. Westfall, MD (1992–1995); William L. White, MD (1992–1995); John Shore, MD (1986–1988); **Radiation Oncologists:** Ardow Ameduri, MD, PA; Carlos J. Hernandez, MD; Anthony J. Campbell, MD (1993–1995); Ejub C. Hadzic, MD (1994–1995); John J. Feldmeier, DO (1988–1992); Mary E. Young, MD, ABR (1990–1992); **Radiation Physicists:** James M. Hevezi, PhD, ABR; Robert G. Waggener, PhD (1987–1995); Edward C. Mok, MSc (1991–1992); **Visual Acuity Examiners:** Leticia G. Curup, COT; Syed Z. Parvez, MD; Steve L. Hand, OD (1990–1992); Karlyn Gansel, OD (1989–1990); **Photographers:** Paul Comeau, CRA; Richard Sabo, CRA (1988–1995); **Ophthalmic Pathologists:** William C. Lloyd III, MD; Andrew Lawton, MD (1986–1995).

Schatz, McDonald, Johnson, and Ai, San Francisco, CA. **Principal Investigator:** Robert N. Johnson, MD; **Clinic Coordinators:** Nubia E. Garcia, RN; Karen Laughlin, LVN; Renee Y. Guertin (1994–1997); Anne Miller (1990–1994); **Ophthalmologist:** Howard Schatz, MD; **Radiation Oncologists:** Sara M. Huang, MD; Adhinand Peddada, MD (1996–1998); John Meyer, MD (1987–1995); Jerold P. Green, MD (1988–1991); **Radiation Physicists:** Avelino C. Dimaya; Devorah Novack, MS; **Visual Acuity Examiners:** Marsha A. Apushkin, MD; Marian M. DiAngelo, RN (1995–1997); Jill Teachout, COA (1989–1990); **Echographers:** Margaret M. Stolarczuk, OD; Patricia M. Wood, LVN; **Photographers:** Irina A. Rozenfeld (Foreign MD); John Uy; Scott M. Wild (1994–1996); Michael McKenzie (1989–1994); Karen Kline (1988–1989).

Ophthalmic Consultants Northwest, Seattle, WA. **Principal Investigator:** Edward B. McLean, MD; **Clinic Coordinators:** Shiela M. Davidson; Pauline M. Manner (1996–1998); Tuija H. Kaarrekoski (1995–1995); Marilyn M. Teeter, MA (1989–1994); **Ophthalmologists:** J. Timothy Heffernan, MD; R. Toby Sutcliffe, MD (1989–1994); **Radiation Oncologist:** James R. Eltringham, MD (1989–1997); **Radiation Physicists:** Juri Eenmaa, PhD; Douglas Schumacher (1989–1994); **Visual Acuity Examiner:** Ralph Archer, COMT (1989–1995); **Photographers:** Robin Wilkins, CRA; Phillip R. Goedert (1995–1995); Christopher Sherman, CRA, RBP (1989–1991); **Ophthalmic Pathologist:** Bruce G. Kulander, MD.

University of Washington, Seattle, WA. **Principal Investigator:** Craig G. Wells, MD; **Clinic Coordinator:** Betty Lawrence, COA; **Ophthalmologists:** Robert E. Kalina, MD; James L. Kinyoun, MD (1990–1995); **Radiation Oncologists:** Wui-Jin Koh, MD; Karen Lindsley, MD; **Radiation Physicists:** Alina Popescu, PhD, CCPM; Karen M. Singer, MS; Mark W. Davidson, MS (1992–1993); Stan Brossard, MS (1990–1992); **Visual Acuity**

Examiner: Yvette Cady, COT (1989–1990); **Photographers:** Brad Clifton; Ron Jones.

Retina Associates of Florida, Tampa, FL. **Principal Investigator:** William S. Grizzard, MD; **Clinic Coordinators:** William J. Malenfant; Janet Traynom, COT; Theresa L. Shannon, COA (1994–1995); Beverly B. Colding, LPN (1994–1994); Mark A. Greenwald (1992–1992); Sandra K. Myers, COT (1989–1992); **Ophthalmologists:** Aaron P. Appiah, MD; Scott M. Friedman, MD; Mark E. Hammer, MD; Peter R. Pavan, MD; Steven M. Cohen, MD (1995–1997); **Radiation Oncologists:** William Assad, MD; Larry Gandle, MD (1992–1995); **Radiation Physicists:** Ignacio A. Ferras III, MS; Ashoka Bhargava, MS, ABR (1993–1996); Zoubir Ouhib, MS (1989–1992); **Visual Acuity Examiner:** Jayne Ann Craig (1995–1995); **Photographer:** Janice M. Lightbourn (1993–1997); **Ophthalmic Pathologist:** Curtis E. Margo, MD; **Following Oncologist:** Curtis C. Adolphson, MD.

Scott and White Memorial Hospital, Temple, TX. **Principal Investigator:** J. Paul Dieckert, MD; **Clinic Coordinators:** Janice M. Jeske, RN; Suzann M. Whatley, COT; Juanice R. Hamm, RN (1987–1995); **Ophthalmologist:** Glen O. Brindley, MD; **Radiation Oncologists:** Alan Y. C. Cheung, MD (1992–1995); Odis D. Skinner, MD (1987–1991); **Radiation Physicist:** Louis H. Deiterman, PhD (1988–1998); **Visual Acuity Examiners:** Terry L. Walters, COT; Donald B. Skelton, COMT (1987–1994); **Echographer:** Vernon M. Hermsen, MD (1988–1989); **Photographers:** George L. Methvin; Barry McMickle (1987–1989).

Ontario Cancer Institute, Toronto, Canada. **Principal Investigators:** E. Rand Simpson, MD; Peter Fitzpatrick, MB, BS (1986–1989); **Clinic Coordinators:** Lee Penney, COA; Andrea Mackie, COA (1987–1994); **Ophthalmologists:** Hesham Lakosha, MD; Hugh McGowan, MD; James H. Oestreicher, MD; Herbert Tanzer, MD; Manijeh Contractor, MD (1994–1998); Andrew Mis, MD (1995–1997); David A. Ewing-Chow, MD (1992–1994); Martin Kazdan, MD (1987–1992); Shyam Mohan R. Kodati, MD, FRC-S(Ed) (1991–1992); John A. McWhae, MD (1990–1991); Katherine Paton, MD (1988–1989); Murray Christianson, MD (1987–1987); **Radiation Oncologists:** Normand J. Laperriere, MD; David Payne, MD; **Radiation Physicists:** Barbara Japp; Ivan Yeung, PhD; Sandi Ahonen, RTT (1989–1993); **Echographer:** Charles J. Pavlin, MD; **Photographers:** Allan Connor; Karen M. Hahn, CRA; Janet Hammond, RN (1989–1991).

Retina Associates Southwest, Tucson, AZ. **Principal Investigator:** Leonard Joffe, MD, FRCS; **Clinic Coordinators:** Mari L. Bunnell, COA; Sally K. Brandon, LPN, COT; Christine Hanson, RN, BSN (1990–1994); Pamela Alcala, COA (1989–1990); Lois J. Loescher, MS, RN (1989–1990); **Ophthalmologists:** Denis M. Carroll, MD; Reid F. Schindler, MD, FRCS(C); **Radiation Oncologists:** James R. Oleson, MD, PhD; Helen Fosmire, MD (1989–1997); J. Robert Cassidy, MD (1989–1995); Terence J. Roberts, MD (1992–1993); Patrick Swift, MD (1990–1990); **Radiation Physicists:** Chee-Wai Cheng, PhD; Bruce Lulu, PhD (1989–1997); Wendell Lutz, PhD (1990–1993); **Visual Acuity Examiners:** Elizabeth S.

Weimann, COT; Darryl C. Freilinger, CRA (1990–1991); **Echographer:** Laura S. Rodgers, COA; **Photographers:** Larry M. Wilson; Stacey A. Halper (1995–1997).

Resource Centers

Chairman's Office, Scheie Eye Institute, Philadelphia, PA. **COMS Group Chairman:** Stuart L. Fine, MD; **Assistants:** Donna DiSciullo; Gretchen E. Goller; Lee McCaffrey, MA (1987–1998); Wendy Tucker (1998); Alissa M. Reich, MEd (1995–1997); Emily Fatula (1993–1995); Virginia Nido (1991–1993); Jane Ebert Torano (1985–1988); **Administrative and Support Personnel:** Carol Lackey (1997); Cheryl Hurley (1991–1997); Debbie Russo (1991–1992).

Coordinating Center: The Wilmer Ophthalmological Institute, Baltimore, MD. **Principal Investigator:** Barbara S. Hawkins, PhD; **Study Statistician:** Marie Diener-West, PhD; **Study Epidemiologist and Protocol Monitor:** Claudia S. Moy, PhD; Jan A. Markowitz, PhD (1985–1991); Diane M. Willis, PhD (1989–1991); **Central Data Coordinators:** René Y. Mobley; Joyce L. Keenan; Townes C. Coates (1989–1997); Wanda S. Jeffreys (1988–1989); Donna C. Optican, MAS (1985–1987); **Clinic Monitors:** Rosemary A. Hollick, MS; Kelly S. Manos, MAS (1991–1998); Christopher Kane, MS, MPH (1994–1995); **Biostatisticians:** Marta J. Marsh, MS; B. Michele Melia, ScM; **Statistical Programmers:** Sandra M. Reynolds, MA; Päivi H. Miskala, MSPH (1996–1998); Kimberly L. King (1994–1996); Craig E. Belongie (1991–1994); Brett A. Hoover (1990–1991); **Systems Analysts and Programmers:** M. Marvin Newhouse; Jonathan S. Kerman (1990–1999); Stephen C. Grubb, MS (1985–1990, 1995–1996); Nimish R. Shah (1994–1995); Paul B. Connor (1988–1992); **Administrative and Support Personnel:** Patricia A. James; Takisha R. Kiah; Lisa A. Lassiter; Nancy A. Prusakowski, MS; Dawn K. Smith; Jeffrey D. Silverman (1998–1999); Irene L. Felicetti (1985–1998); Darren T. Otero, MM (1995–1998); Katasha Smart-Davis (1992–1994); George T. Allen (1989–1993); Juanchulla Terry (1990–1992); Darryl M. Harper (1989–1991); Amanda Davis (1987–1990).

Echography Center: Mars Hill College, Mars Hill, NC. **Principal Investigator:** Sandra Frazier Byrne; **Coordinators:** Valerie Bruns; Linda Pritchard (1996–1999); Florence Zerbarini (1994–1996); Jeanette Moreno (1992–1994); Pat Superfine-Rivera (1987–1992); Michelle Comella (1986–1987); **Graders:** E. Kym Gendron (1985–1994); J. Randall Huges (1985–1994); Maria Rivera (1993–1994); Cathy DiBernardo, RN (1985–1991).

Pathology Center: University of Wisconsin, Madison, WI. **Principal Investigator:** Daniel M. Albert, MD; **Coordinators:** Michelle Zimbric; Tracy Perkins, MPH, COA (1998); Nancy L. Robinson (1986–1998); **Ophthalmic Pathologists:** Guillermo de Venecia, MD (1986–1999); Ingolf Wallow, MD (1986–1998); Thaddeus Dryja, MD (1986–1992); **Technical and Support Personnel:** Lorie Buckley (1992–1994); Emily Hajar (1989–1992); Bartholomew Johnson (1989–1992).

Photograph Reading Center: The University of Iowa

Hospitals and Clinics, Iowa City, IA. **Principal Investigator:** H. Culver Boldt, MD; Thomas A. Weingeist, MD, PhD (1986–1999); **Computer Consultant:** Paul R. Montague; **Coordinators:** Diane Anderson; Laurie Smith (1988–1992); Judy Strauss (1986–1988); **Ophthalmologists:** Edwin M. Stone, MD, PhD (1993–1999); James C. Folk, MD (1988–1992); **Photograph Readers:** Carolyn Vogel; Ivy Andersen (1997–1998); William Ward (1994–1996); Christine Taylor (1991–1993); Luan Streb (1989–1992); Ray Northway (1986–1991).

Radiological Physics Center: M. D. Anderson Hospital, Houston, TX. **Principal Investigator:** William F. Hanson, PhD; **Physicists:** David Followill, PhD; Connel Chu, MS (1995–1996); Thomas Kirby, PhD (1986–1988); Ramesh C. Tailor, PhD (1992–1994); Carter B. Schroy, PhD (1989–1991); **Technical Support Personnel:** Elizabeth Engelberg; Paula Kennedy (1986–1996); Cathy Kasper (1987–1991).

Sponsor

National Eye Institute, Bethesda, MD. **Program Staff:** Natalie Kurinij, PhD.

Committees as of December 31, 1998

Steering Committee. Chairman: Stuart L. Fine, MD; **Members:** John D. Earle, MD; Barbara S. Hawkins, PhD; Natalie Kurinij, PhD; Bradley R. Straatsma, MD.

Executive and Quality Assurance Committee. Chairman: Stuart L. Fine, MD; **Co-Chairmen:** John D. Earle, MD; Bradley R. Straatsma, MD; **Ex Officio Members:** Daniel M. Albert, MD; Sandra Frazier Byrne; Marie Diener-West, PhD; William F. Hanson, PhD; Barbara S. Hawkins, PhD; Rosemary A. Hollick, MS; Marta J. Marsh, MS; B. Michele Melia, ScM; Renee Y. Mobley; John M. Kirkwood, MD; Natalie Kurinij, PhD; Claudia S. Moy, PhD; David Payne, MD; Dennis M. Robertson, MD; Andrew P. Schachat, MD; Thomas A. Weingeist, MD, PhD; **Other Members:** Mike Hartnett, COT; Robert Caldwell, P.A.; Beryl McCormick, MD.

Data and Safety Monitoring Committee. Chairman: Matthew D. Davis, MD; **Voting Members:** Marc S. Ernst-off, MD; Frederick L. Ferris III, MD; Laël C. Gatewood, PhD; Robert J. Levine, MD; Robert W. Makuch, PhD; James Marks, MD; Travis A. Meredith, MD; Donald L. Patrick, PhD, MSPH; **Ex Officio Non-Voting Members:** Marie Diener-West, PhD; John D. Earle, MD; Stuart L. Fine, MD; Barbara S. Hawkins, PhD; Natalie Kurinij, PhD; Claudia S. Moy, PhD.

Radiation Therapy and Physics Committee. Chairman: John D. Earle, MD; **Members:** William F. Hanson, PhD; Robert W. Kline, PhD; Beryl McCormick, MD; Karl R. Olsen, MD; Marvin Rotman, MD.

Pathology Review Committee. Chairman: Daniel M. Albert, MD; **Members:** W. Richard Green, MD; Hans Grossniklaus, MD.

Mortality Coding Committee. Chairman: James K. V. Willson, MD; **Members:** Daniel M. Albert, MD; Robert E. Scully, MD; **Ex Officio Members:** Marie Diener-West, PhD; Lee McCaffrey, MA; Claudia S. Moy, PhD.